

EmoPhone: A Multi-Wave Dataset for In-the-Wild Mobile and Wearable Affect Sensing

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Abstract

We introduce a three-wave, in-the-wild multimodal dataset for affect sensing that integrates smartphone sensing, wearable sensing, and dense experience-sampling-method (ESM) labels collected annually from 2020 to 2022. The dataset supports moment-level affect modeling through a shared dimensional label core across all waves, with additional affective descriptors available in the third wave (D-3). We describe the resource in terms of study design, temporal density of in-situ labels, and sensing and label coverage across waves. To support evaluation within this resource, we define an initial three-setting benchmark spanning temporal prediction from within-user history, within-wave cross-user generalization, and cross-wave generalization in which each wave is treated as a separate dataset. Our benchmark results show that the strongest method family depends on the evaluation setting: supervised baselines perform best in the temporal setting, unsupervised domain adaptation is strongest overall in the within-wave cross-user setting, and domain generalization shows the strongest overall cross-wave performance, although its margin over strong baselines is modest. These findings indicate that robust mobile affective computing is constrained not only by label availability but also by substantial participant-level variability and realistic cross-wave differences inherent in longitudinal in-situ deployments.

1 Introduction

Passive sensing from smartphones and wearables has created new opportunities for modeling human states and behaviors in everyday life [49, 39, 53, 14, 42, 6]. Recent longitudinal mobile sensing benchmarks have shown the value of multi-year collection for studying robustness and generalization, but their benchmark targets are typically broad behavioral or mental-health outcomes collected at relatively sparse intervals.

Existing mobile-sensing resources typically emphasize either long-term behavioral and mental-health outcomes, large-scale in-situ labels, or generalization benchmarking, but rarely combine dense moment-level affect labels, smartphone-plus-wearable sensing, multi-wave collection, and

a structured benchmark in one resource. This gap matters because affective states are inherently dynamic, short-horizon, and person-dependent [31, 17].

Our resource is tailored for this longitudinal setting: a three-wave, in-the-wild dataset (D-1, D-2, and D-3) integrating smartphone sensing, wearable sensing, and dense in-situ affective labels collected via an in-situ experience sampling method (ESM) [13]. Across waves, valence, arousal, stress, and disturbance form a shared benchmarkable label core, while D-3 adds an expanded PANAS-style affect inventory [52] for finer-grained emotion analysis. The combination of multi-year coverage, naturalistic Android-phone and wearable sensing, and a relatively homogeneous cohort provides a cleaner setting for studying cross-user and cross-wave variation [53].

We pair the dataset release with a characterization-first benchmark that can be applied to other datasets. After documenting temporal density, sensing and label coverage, participation, and participant/wave heterogeneity, we evaluate three increasingly demanding settings: within-user temporal prediction, within-wave generalization to unseen users, and cross-wave generalization to held-out waves. Together, these settings separate basic temporal learnability from user-level and wave-level robustness, providing a foundation for studying affect prediction, personalization, and robust generalization in in-the-wild mobile affective computing.

2 Related Work & Positioning

Longitudinal mobile sensing and affect benchmarks: Recent longitudinal mobile sensing benchmarks emphasize evaluation under realistic distribution shift rather than only within-dataset prediction [55, 33, 30]. GLOBEM is the closest benchmark-style precedent: it releases multi-year mobile and wearable sensing datasets and evaluates cross-dataset generalization for depression-related behavior modeling [54]. Its targets, however, are sparse weekly or end-of-term mental-health outcomes, whereas EmoPhone focuses on dense moment-level affect labels. K-EmoPhone is the closest single-wave predecessor, providing real-world smartphone and wearable sensing with in-situ emotion, stress, and attention labels [20]. Rather than extending K-EmoPhone itself, we introduce three newly collected dataset waves and benchmark settings for user- and wave-level generalization, with denser labels and richer D-3 affect descriptors. DiversityOne, the College Experience Study, and BRIGHTEN provide important daily-life or remote mobile-sensing resources [4, 34, 37], but they do not combine smartphone-plus-wearable sensing, dense in-situ affect labels, a shared multi-wave benchmark, and richer D-3 affect descriptors. FEEL further motivates our heterogeneity-aware benchmarking framework by showing that experimental setting, device type, and labeling strategy shape generalization in physiological emotion recognition [45]. Table 1 summarizes these differences.

3 EmoPhone Dataset: Design and Characterization

This section describes the three-wave EmoPhone resource, its study protocol, released data, temporal label density, and sensing/label coverage. We use descriptive analyses here to clarify what the dataset contains and what kinds of benchmark claims it can support. We reserve the term “validation” for the benchmark evaluation in later sections, where models are tested under temporal, cross-user, and cross-wave settings.

Table 1: **Related datasets.** GLOBEM = closest longitudinal benchmark; DiversityOne = closest problem comparator; K-EmoPhone = closest single-wave predecessor. EmoPhone uniquely combines multi-wave benchmarking, mobile+wearable sensing, dense affect labels, homogeneous cohort, and richer D-3 descriptors.

Dataset	Bench.	Users	Labels	Multi-yr	Sources	Density	Homogeneity	Rich Affect
GLOBEM	Yes	497	~10–12	Yes	Mobile+Wear	Low	High	No
DiversityOne	No	782	~150–200	No	Mobile	High	Low	No
College Exp.	No	215	~100–150	Yes	Mobile	Low	High	No
BRIGHTEN	No	~2,193*	Var.	No	Mobile	Low	Low	No
K-EmoPhone	No	77	~70	No	Mobile+Wear	High	High	No
EmoPhone (ours)	Yes	328	~100/200	Yes	Mobile+Wear	High	High	Yes

* BRIGHTEN: severe remote attrition reduced dense subset.

Table 2: Overview of dataset characteristics across the three collection waves. Active day denotes a day with at least one valid ESM response.

Characteristic	D-1 (2020)	D-2 (2021)	D-3 (2022)	Total / Notes
<i>Demographics</i>				
Recruited (N)	102	112	114	328
Final sample (N)	92	99	106	297
Gender (M/F)	64/28	75/24	67/39	206/91
Mean age, SD	22.9 (3.4)	24.0 (3.6)	23.6 (3.5)	—
Psych. baselines	PSS-10, GHQ-12, PHQ-9	PSS-10, GHQ-12, PHQ-9	PSS-10, GHQ-12	PHQ-9 absent in D#3
<i>Data Collection</i>				
Nominal study window	~4 weeks	~4–5 weeks	~4–5 weeks	participant span varies
Mean span (days)/person	22.51	31.67	30.18	realized observation span
Mean active days/person	17.97	30.36	29.66	active day = ≥ 1 valid ESM
ESM sent freq. (per day)	15	12	10	↓ intentional reduction
Avg. ESM responses per active day	6.70	6.99	7.03	computed over active days only
Total ESM responses	10,259	21,042	21,838	53,139
ESM avg/person, SD	111.51 (51.09)	212.55 (44.05)	206.02 (24.72)	—
<i>Sensors & Instruments</i>				
Smartphone sensing	✓	✓	✓	Continuous
Wearable (wrist)	Fitbit HR	Fitbit HR	Fitbit HR	Continuous
Keystroke sensing	—	✓	✓	Added from D#2 onward

3.1 Three-wave resource overview

As shown in Table 2, our resource consists of three longitudinal, in-the-wild multimodal datasets, denoted D-1, D-2, and D-3, collected annually from 2020 to 2022. Each wave was conducted with university students over approximately four weeks and combined repeated experience-sampling labels with passive sensing from participants’ own Android phones and wearable devices. The three waves share a common collection backbone while differing in several wave-specific design choices, enabling both unified benchmarking and analysis of cross-wave variation.

The resource is substantial in both participant count and label volume. D-1, D-2, and D-3 include 102, 112, and 114 recruited participants, with 92, 99, and 106 retained after quality control based on ESM response rate and smartphone/wearable data coverage thresholds (Appendix A). The corresponding ESM totals are 10,259, 21,042, and 21,838, with mean per-participant counts of 111.51, 212.55, and 206.02. Because participant observation windows vary across waves, we report ESM frequency over active days rather than over a fixed calendar window: D-1, D-2, and D-3 yield 6.70, 6.99, and 7.03 responses per active day, respectively. These values show that the resource is not only multi-wave but also densely labeled at the moment level.

The waves remain comparable enough for shared-label benchmarking while preserving meaningful cross-wave heterogeneity. Across D-1, D-2, and D-3, the resource differs in questionnaire structure, sensing availability, and label coverage. For example, keystroke sensing is available from D-2 onward, and D-3 includes additional affective descriptors beyond the shared core labels (valence, arousal, stress, disturbance), including Happy, Relaxed, Cheerful, Content, Sad, Angry, Anxious, and Depressed. This structure makes the dataset comparable enough for shared-label benchmarking while preserving meaningful wave-level heterogeneity. This label design makes the resource relevant to a range of tasks in affective computing, including emotion recognition, stress detection, and attention-related modeling [35, 20].

3.2 Study protocol and released data

The study protocol was designed to capture affective states in naturalistic daily life rather than under laboratory control. Participants were recruited from the same university-student population under an Institutional Review Board (IRB)-approved protocol, provided informed consent, installed ABC Logger [20] on their own Android phones, and wore the designated Fitbit device during the study period. This cohort design reduces some broad demographic and lifestyle confounds, making cross-user and cross-wave differences easier to interpret, while still retaining substantial participant-level variability.

Across waves, the study collected three main streams during daily life: repeated ESM prompts for momentary affective and cognitive states, passive smartphone sensing through ABC Logger [20],

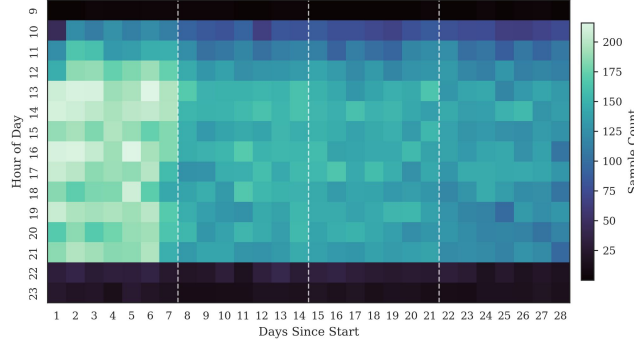


Figure 1: Temporal distribution of ESM labels across D-1, D-2, and D-3 by day since study start and hour of day. Labels concentrate during waking hours while remaining distributed across the collection period, supporting sustained moment-level supervision.

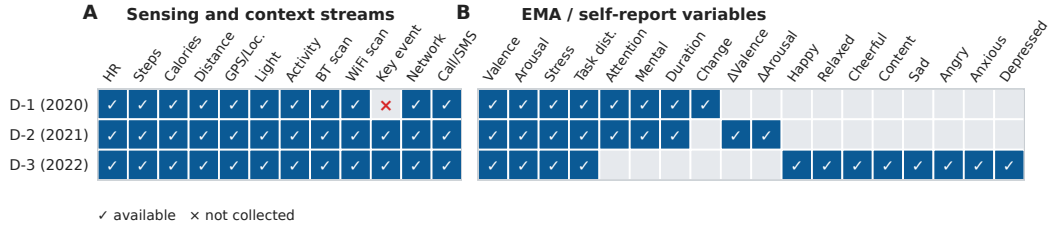


Figure 2: Sensor and label coverage across waves. A) Sensor and context streams, including shared smartphone/Fitbit streams and wave-specific additions such as key stroke events from D-2 onward. B) ESM/self-report variables, with valence, arousal, stress, and task disturbance forming the shared benchmark label core, and D-3 adding richer affect descriptors (e.g., happy, angry, depressed).

and wearable physiological and mobility streams from Fitbit HR. Participant-level metadata, including demographics, personality traits, and psychological scales, were also collected to support cohort characterization and secondary analyses. The released resource includes these ESM labels, smartphone-derived behavioral and contextual signals, wearable streams, and participant-level metadata; detailed file schemas, sensor fields, preprocessing notes, and release structure are provided in Appendix A.

3.3 Temporal density of in-situ affect labels

Figure 1 complements the summary statistics in Table 2 by showing when ESM labels are observed over the study horizon. Responses are distributed across many study days with clear daytime concentration, indicating sustained engagement rather than short bursts of labeling, and supporting the temporal and cross-user evaluations used below.

3.4 Sensing and label coverage across waves

Figure 2 summarizes the protocol-level structure of the resource. At the sensing level, the waves share a common backbone to support harmonized evaluation, while still preserving realistic cross-wave heterogeneity. At the label level, the dataset combines a shared benchmarkable core with wave-specific enrichment. This dual structure is central to the later benchmark design: the shared sensing and label backbone supports common evaluation across waves, whereas the wave-specific sensing and richer D-3 descriptors make cross-wave generalization and fine-grained affective analysis meaningfully challenging.

Table 3: Overview of the three benchmark settings. Full model inventories and hyperparameter spaces are reported in Table 21 in Appendix C.2.

Benchmark	Evaluation setting	Targets	Domain / adaptation protocol	Model families
Temporal	Per-user chronological 60/20/20 train/validation/test split within the first 30 days; users are excluded if any split is empty or lacks both classes.	Shared labels in D-1/D-2/D-3; D-3-specific affect labels in D-3.	Within-user prediction; no domain shift is introduced by design.	Baselines only
Within-wave	User-grouped stratified 5-fold evaluation within each wave; no user overlap between train and test folds; per-user sensor-feature normalization.	Shared labels in D-1/D-2/D-3; D-3-specific affect labels in D-3.	User identity defines DG domains. UDA uses unlabeled held-out-user features but no target labels.	Baselines, DG, UDA
Cross-wave	Leave-one-dataset-out evaluation with both single-source→single-target and two-source→one-target transfer; features are aligned by canonicalized cross-wave intersection.	Shared labels only, so source and target label spaces remain aligned.	Train/test split is wave-level. For DG, each source-wave participant is treated as a separate domain. UDA uses unlabeled target-wave features but no target labels.	Baselines, DG, UDA

4 Benchmark framework: tasks and experimental design

4.1 Motivation: cross-user and cross-wave heterogeneity

The dataset characterization above shows that EmoPhone has both a shared backbone and realistic heterogeneity. The shared sensing and label structure enables common evaluation across waves, while participant-level and wave-level differences make random sample-level evaluation insufficient. We therefore design the benchmark around three settings that progressively separate temporal learnability, cross-user generalization, and cross-wave robustness.

4.2 Benchmark overview

We organize the benchmark as an initial three-setting evaluation suite (temporal, within-wave, and cross-wave) that reflects increasingly demanding forms of generalization, beginning with the prediction of a user’s own earlier history, moving to generalization between unseen users within the same wave, and culminating in generalization between waves treated as distinct datasets [45, 54]. This structure allows the benchmark to separate basic learnability from user-level generalization difficulty and from the stricter challenge of cross-wave robustness.

Accordingly, the *temporal* benchmark asks whether affective states are predictable from a user’s own earlier observations, the *within-wave* benchmark asks whether models trained on some users generalize to unseen users within the same wave, and the *cross-wave* benchmark asks whether models trained on one or more waves generalize to a held-out wave under a shared feature and label space. Framing the benchmark this way makes clear that the three settings are complementary components of one evaluation suite rather than three disconnected experiments. Table 3 summarizes the evaluation settings and model families.

All benchmark features are extracted from sensor records observed at or before the corresponding ESM prompt time. Appendix A.7 details the active feature windows, point-in-time features, and missing-data handling rules.

4.3 Prediction targets

The benchmark is built around a shared core label space consisting of valence, arousal, stress, and disturbance, because these four targets can be aligned consistently across D-1, D-2, and D-3 and therefore support unified comparison across waves. Consequently, they serve as the common evaluation targets whenever cross-wave comparability is required.

In addition to this common label layer, D-3 provides a rich semantic descriptor layer consisting of happy, relaxed, cheerful, content, sad, anxious, depressed, and angry, which enables a more semantically fine-grained view of affect than the shared dimensional labels alone. These richer targets are valuable because they allow the benchmark to examine whether some affective concepts are easier to predict or generalize than others once the label space moves beyond the common core.

The benchmark targets follow directly from this design. The temporal and within-wave benchmarks evaluate the shared labels across all applicable waves, and they also include the richer D-3-specific labels when those targets are available. The cross-wave benchmark, on the contrary, is restricted to the shared labels so that comparison across datasets remains strictly aligned. This design preserves a common label space for cross-wave evaluation while still allowing richer semantic analysis where the label set supports it.

For binary classification, continuous ESM responses are converted into user-relative HIGH/LOW targets using a user-specific mean split after scale harmonization; Appendix A.7.6 reports the rescaling and thresholding details.

4.4 Temporal benchmark

The temporal benchmark measures within-user predictability by testing whether future affective states can be predicted from earlier self-observations. It verifies that the dataset contains temporal signal and serves as a reference point for measuring performance drops under generalization.

Task construction: Tasks are defined at the (label, dataset) level. We evaluate the four shared labels in each wave (D-1, D-2, and D-3) as separate binary classification problems; in D-3, we additionally evaluate the eight affect labels as D-3-only tasks.

Splitting: For each user, we consider the first 30 days of samples, order them chronologically, and split them into 60%/20%/20% train/validation/test partitions; the middle 20% serves as the temporal validation slice. We then concatenate splits across users while preserving temporal order within each user.

Models: We evaluate the supervised baseline family, which includes both tree-based methods (XGBoost [5], LightGBM [22]) and basic and tabular neural networks (MLP, ResNet [12], TabNet [2], DCN [51], SAINT [46], TabTransformer [15], FTTransformer [10]). We do not include domain generalization (DG) or domain adaptation (DA) methods in this benchmark because source and target distributions coincide by construction.

4.5 Within-wave benchmark

The within-wave benchmark evaluates user-independent generalization within each wave. Models must generalize from seen users to unseen users while the wave-specific collection protocol is held fixed, isolating person-level heterogeneity as the sole source of distribution shift.

Task construction: Within-wave tasks are again defined at the level (label, dataset). The four shared labels are evaluated in each of D-1, D-2, and D-3, and the eight D-3-only affect labels are evaluated in D-3 only. Per-user z -score normalization is applied to reduce cross-user scale effects before training (details in Appendix C.1.3). Feature normalization statistics are fitted only on the training split and then applied to validation/test users to avoid data leakage.

Splitting and preprocessing: We use stratified group 5-fold cross-validation with user identity as the grouping variable so that each user’s samples appear in exactly one fold. Appendix C.1.2 provides the exact split definition.

Models: We compare three method families under a shared training loop, the temporal-benchmark baselines, DG methods (user as domain), and UDA methods. In the within-wave setting, the domain variable is user identity; Appendix C.2.1 details how domain labels are instantiated for DG/UDA methods.

4.6 Cross-wave benchmark

The cross-wave benchmark evaluates cross-wave generalization by treating each wave as a distinct dataset. By combining unseen-user heterogeneity with shifts in study conditions, feature availability, and label distributions, this setting is the strictest one and is intended to approximate real-world dataset shift.

Task construction: We follow a leave-one-dataset-out protocol, with each wave taking a turn as the target. This yields six one-to-one generalization directions (e.g. D-1 $\rightarrow$ D-2) and three two-to-one

directions (e.g. D-1+D-2→D-3). The cross-wave benchmark is restricted to the four shared labels so that source and target label spaces are aligned.

Splitting and preprocessing: For each generalization direction, we train on the source wave(s) and evaluate on the held-out target wave. We align feature spaces using a fixed three-way intersection across D-1/D-2/D-3; Appendix C.1.1 details the alignment procedure.

Models: As in the within-wave benchmark, we compare baseline models, DG methods, and unsupervised domain adaptation (UDA) methods. The cross-wave split is defined at the wave level: models are trained on one or more source waves and evaluated on a held-out target wave. For DG training, however, domains are instantiated at the participant level, using each source-wave participant as a separate domain. Thus, in both single-source and multi-source cross-wave settings, DG methods learn from multiple user domains within the source training pool. UDA methods use unlabeled target-wave features during adaptation, but target labels are never used for training, validation, or model selection.

4.7 Cross-method comparability

The within-wave and cross-wave benchmarks compare three method families (baselines, DG, and UDA), so the three sources of unfairness in such comparisons are explicitly controlled. The corresponding values (optimizer, loss, batch size, schedule, seeds, and per-method search spaces) are documented in Appendix C.2.3 and Tables 22–24.

Data treatment. ID-linked and temporal identifiers (participant IDs, timestamps) are excluded from the feature set so that user identity cannot leak into the inputs; standardization statistics are computed only on the training split; and multi-source training pools are aligned via feature-space intersection rather than padding (Appendix C.1). These rules apply uniformly to every method, so no model is given a different feature set or split.

Capacity matching. All DG and UDA methods share the same MLP encoder as the MLP baseline. Performance gaps between families therefore reflect the training objective and adaptation strategy rather than backbone capacity.

Tuning budget. Every method is tuned with 30 Optuna [1] trials per (benchmark, label, dataset, model) configuration, selected on validation AUROC, and trained under a single shared training loop and schedule, so no method is favored or penalized by under-tuning or schedule mismatch. Concretely, all neural methods use AdamW with cross-entropy, batch size 512, up to 50 epochs with patience 20 on validation AUROC, and seed 42; the full per-method search spaces are listed in Appendix C.2.3.

5 Benchmark Results

We summarize the main results in this section, with full details in Appendix C. We report results for the three benchmark settings (temporal, within-wave, and cross-wave) and then interpret them as a label-dependent difficulty ladder.

5.1 Within-User Temporal Prediction

The temporal benchmark establishes a within-user predictability reference for each target (Table 4; the full model-by-(label, dataset) matrix is provided in Appendix C). Predictability is strongly label-dependent: on the shared labels, stress is the most predictable (mean AUROC 0.628 ± 0.016), followed by arousal (0.609 ± 0.035) and valence (0.601 ± 0.031), whereas disturbance lies closest to chance (0.555 ± 0.005). The D-3-only affect labels span a comparably wide range, indicating that within-user signal varies substantially across affect dimensions even when training and test distributions coincide.

5.2 Within-Wave Cross-User Generalization

The within-wave benchmark tests cross-user generalization via a stratified 5-fold group split by participant. Table 5 summarizes family-level results on the shared labels, and Appendix C reports full per-task matrices and DG/UDA breakdowns.

Table 4: Within-user temporal benchmark summary. Shared labels report the best model and mean test AUROC $\pm$ std across D-1/D-2/D-3; D-3-only labels report the best model and AUROC on D-3. Full results are in Appendix C (Table 25).

Label	Best Model	Test AUROC
Shared labels (mean across datasets)		
Stress	XGB	0.627 ± 0.016
Arousal	XGB	0.609 ± 0.035
Valence	XGB	0.601 ± 0.031
Disturbance	SAINT	0.555 ± 0.005
Mean $\pm$ Std across shared labels	–	0.598 ± 0.031
D-3 rich labels		
Angry	XGB	0.673
Sad	LGB	0.607
Content	XGB	0.601
Cheerful	LGB	0.599
Happy	LGB	0.593
Relaxed	XGB	0.585
Anxious	SAINT	0.575
Depressed	FTTransformer	0.548

Table 5: Family-best shared-label AUROC for within-wave and cross-wave benchmarks. Each cell reports the best model within a family, averaged across datasets or transfer directions. Full per-model results are in Appendix C.

Label	Within-wave			Cross-wave		
	Baseline	DG	DA	Baseline	DG	DA
Stress	0.544	0.554	0.554	0.525	0.525	0.522
Arousal	0.576	0.574	0.587	0.552	0.564	0.538
Valence	0.560	0.561	0.577	0.540	0.541	0.525
Disturbance	0.543	0.550	0.563	0.535	0.537	0.532
Mean $\pm$ Std	0.556 ± 0.016	0.560 ± 0.011	0.570 ± 0.015	0.538 ± 0.011	0.542 ± 0.016	0.529 ± 0.007

UDA is strongest under cross-user shift. On the four shared-label tasks, UDA leads on average with mean AUROC 0.570 versus 0.560 (DG) and 0.556 (baseline); UDA is best-performing on arousal, valence, and disturbance, and is tied with DG on stress (both 0.554). The same pattern largely holds on the eight D-3-only affect labels, where UDA is best-performing on seven of the eight targets; baselines remain competitive on individual targets but are not consistently strongest when generalizing to unseen users.

No single UDA method dominates. The advantage of UDA is shared across objectives rather than driven by one method: as shown in Tables 6 and 30, the best UDA model differs from one shared label to the next, and the same is true on the D-3-only affect labels. The UDA family as a whole, rather than any single algorithm, is what carries the cross-user-shift advantage.

5.3 Cross-Wave Generalization

The cross-wave benchmark evaluates one-to-one and two-to-one generalization across waves with feature spaces aligned by intersection (Table 5; per-direction results in Appendix C). Family-level differences narrow sharply relative to the within-wave setting, with baseline, DG, and UDA means falling within a 0.013 AUROC band (0.538, 0.542, and 0.529). Although DG attains the highest family-level AUROC on every shared label, the best individual DG method differs by label (MASF, IRM, Fish, and CSD in Table 6), and its margin over supervised baselines remains small. UDA, in contrast, loses 0.041 AUROC on average relative to the within-wave benchmark, indicating that user-level adaptation does not transfer well across wave-level shifts; performance also varies considerably by source-target direction, consistent with heterogeneous mixtures of covariate and label shift across wave pairs.

5.4 Benchmark Ladder

Taken together, the three benchmarks describe a label-dependent difficulty ladder rather than a single ranking (Table 6). Averaged across the four shared labels, the temporal setting performs best (0.598),

Table 6: Best-achieved shared-label AUROC across benchmarks. Values are mean $\pm$ std across datasets for temporal/within-wave settings and across nine transfer directions for cross-wave.

Label	Temporal		Within-wave		Cross-wave	
	Model	AUROC	Model	AUROC	Model	AUROC
Stress	XGB	0.628 ± 0.016	SHOT	0.554 ± 0.018	MASF	0.525 ± 0.013
Arousal	XGB	0.609 ± 0.035	MCD	0.584 ± 0.022	IRM	0.564 ± 0.020
Valence	XGB	0.601 ± 0.031	MCC	0.576 ± 0.010	Fish	0.541 ± 0.008
Disturbance	SAINT	0.555 ± 0.005	SHOT	0.558 ± 0.009	CSD	0.537 ± 0.012
Mean $\pm$ Std	–	0.598 ± 0.031	–	0.568 ± 0.014	–	0.542 ± 0.016

the within-wave setting drops to 0.568, and the cross-wave setting drops further to 0.542 and is the hardest in aggregate. The drop is uneven across labels: stress shows the biggest fall from temporal to cross-wave (0.103 AUROC), followed by valence (0.060) and arousal (0.045), while disturbance remains within a narrow band (0.537–0.558 across all three settings). No single setting therefore characterizes difficulty across all shared targets.

6 Discussion

6.1 Implications of the benchmark findings

The three-setting benchmark shows that benchmark conclusions depend strongly on the evaluation setting. Temporal prediction establishes that several affect labels are learnable from a user’s own history, while within-wave evaluation shows that UDA methods benefit from unlabeled held-out-user features under cross-user shift. Cross-wave evaluation is harder and more mixed: DG achieves the strongest aggregate family-level result, but only by a modest margin over supervised baselines, and UDA no longer provides the same advantage. These results indicate that user-level adaptation does not automatically translate into wave-level robustness, and that no single score adequately summarizes performance across within-user, cross-user, and cross-wave settings.

6.2 Recommended use cases

This resource is well suited to moment-level affect prediction from passive sensing, user-aware or personalized affect modeling, across-user generalization within a cohort, and cross-dataset generalization studies involving baseline, DG, and UDA methods [20, 55, 45, 30]. It is also a useful foundation for richer affect representation learning, especially when models need to connect broad affective dimensions such as valence and arousal with more specific descriptors available in D-3 [16, 48]. For users primarily interested in robust method comparison, the within-wave and cross-wave benchmarks are the most informative settings. For users interested in target construction or label semantics, D-3 provides the richest starting point.

6.3 Limitations

The dataset should also be read with clear but proportionate boundaries. First, the cohort is student-centered and relatively homogeneous in age and life context, which is useful for reducing some confounds but limits broader population coverage. Second, the sensing design is Android-only, so the release does not directly represent iOS-based usage patterns. Third, the three waves are not perfectly identical in sensing and questionnaire design, which improves the resource in some respects but also introduces real cross-wave differences that affect direct comparability. Fourth, modality overlap is strong but not complete: some streams are partial or wave-specific. Fifth, user-relative label binarization makes HIGH/LOW targets reflect within-person deviations rather than absolute affect severity, so they should not be interpreted as clinical or population-level affect measures. Finally, the richer PANAS-style affect labels appear only in D-3, so fine-grained emotion analyses cannot yet be carried out symmetrically across all three waves. These are not flaws to hide; they are the practical boundaries that define what kinds of claims the dataset can support most cleanly.

Accordingly, the benchmark should be interpreted as a scoped evaluation suite for the EmoPhone three-wave resource, not as a definitive benchmark for all populations, device ecosystems, or mobile affect-sensing deployments.

7 Conclusion

We presented a three-wave, in-the-wild multimodal resource for affective sensing that combines smartphone sensing, wearable sensing, and dense ESM labels collected from 2020 to 2022. By pairing a shared dimensional label core with a richer semantic descriptor layer in D-3, the resource provides a rigorous foundation for evaluating robustness under increasingly challenging generalization settings. Our initial benchmark results show that method rankings vary across evaluation settings. Taken together, these findings suggest that the central bottleneck in mobile affective sensing is not merely label density, but the combination of substantial user-level variability and realistic cross-wave differences that emerge in multi-year deployments. More broadly, this work offers a characterization-first template for future dataset design in mobile affective computing.

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A Study Documents

A.1 Study Documents

We provide the following documents in the supplementary folder:

- **OT Materials (Original, Korean)** – The original participant orientation materials administered in Korean during the onboarding session.
- **OT Materials (English Translation)** – An English translation of the participant orientation materials provided for reviewer accessibility.
- **Informed Consent Form (English Translation)** – An English translation of the informed consent form shown to participants before enrollment.

A.2 Study Demographics

Table 7 provides the full per-wave participant and data summary. Counts are reported both for all recruited participants and for the analysis-eligible subset after quality-control (QC) filtering. QC excluded participants with wearable data coverage below 50%, smartphone data coverage below 50%, or an ESM response rate below 30%.

Table 7: Per-wave participant and data summary.

Characteristic	D-1 (2020)	D-2 (2021)	D-3 (2022)	Total / Notes
<i>Demographics</i>				
Recruited (<i>N</i>)	102	112	114	328
Final sample (<i>N</i>)	92	99	106	297
Gender (M/F)	64/28	75/24	67/39	206/91
Mean age, SD	22.9 (3.4)	24.0 (3.6)	23.6 (3.5)	—
<i>Data Collection</i>				
Nominal study window	~4 weeks	~4–5 weeks	~4–5 weeks	participant span varies
Mean span (days)/person	22.51	31.67	30.18	realized observation span
Mean active days/person	17.97	30.36	29.66	active day = ≥ 1 valid ESM
ESM sent freq. (per day)	15	12	10	↓ intentional reduction
Avg. ESM responses per active day	6.70	6.99	7.03	computed over active days only
Total ESM responses	10,259	21,042	21,838	53,139
ESM avg/person, SD	111.51 (51.09)	212.55 (44.05)	206.02 (24.72)	—

All three cohorts were drawn from the same university student population, resulting in broadly comparable age distributions across waves. The female proportion ranges from 23% to 37% across waves after QC. The PSS-10 and GHQ-12 psychological baseline measures are available for all three waves and are the recommended baseline psychological covariates for cross-wave analyses. Nominal study window rows in Table 7 report the overall recruitment and data-collection span for each wave.

A.3 Data Collection Infrastructure and Cross-Wave Design Differences

A.3.1 Smartphone Data Collection Application

Passive sensor data and ESM prompts were collected via ABC Logger, a custom Android application deployed on participants’ personal devices. The application runs continuously in the background,

logging sensor streams and delivering randomised ESM prompts within a defined daily window. Data were uploaded to a remote server hourly in D-2 and D-3, and via periodic background batches in D-1. Participants were required to keep the application and all wearable devices active for at least 12 hours per day.

Android version requirements differed across waves: Android ≥ 7.0 was required in D-1, and Android ≥ 8.0 was required in D-2 and D-3. The research team monitored each participant’s upload status and ESM compliance every 2–3 days throughout each collection period. Participants who were not correctly uploading data, or who fell below a 50% ESM response rate, were individually notified and prompted to supplement their responses.

A notable different protocol design was introduced in D-2: a participant-facing progress counter was added to the application UI, allowing participants to track their own daily ESM completion status. This substantially improved response rates without requiring researcher-initiated reminders. After the collection period, participants returned all wearable sensors, uninstalled the application, and completed a set of post-study psychological questionnaires.

A.3.2 Wearable Devices

A single wearable device was used across all three waves.

Fitbit Inspire HR – a wrist-worn device collecting heart rate (1 Hz), step count, calorie burn, and distance (all at 1/60 Hz). Used continuously throughout all three waves for all participants.

Table 8: Wearable device availability per wave.

Device	Type	Waves	Coverage Notes
Fitbit Inspire HR	Wrist-worn	D-1, D-2, D-3	Continuous throughout study period

A.3.3 Cross-Wave Design Differences

Although the three waves share a common collection backbone, the three waves differ in several collection parameters. Table 9 summarises the key protocol parameters per wave, followed by the Key differences across waves.

Table 9: Cross-wave protocol parameters.

Parameter	D-1	D-2	D-3
Android requirement	≥ 7.0	≥ 8.0	≥ 8.0
ESM prompts / day	15 (random)	12 (random)	10 (random)
Data upload cadence	Periodic batches	Hourly	Hourly
Keystroke sensing	×	✓	✓
Participant-facing progress counter	×	✓	✓
ESM questionnaire items	8	9	12

Key differences across waves:

- **Keystroke sensing added in D-2.** KeyEvent data was introduced to capture fine-grained smartphone interaction dynamics – typing speed, inter-key intervals, and keyboard type – as potential behavioural indicators of cognitive and emotional state.
- **ESM prompt frequency reduced across waves.** The intentional reduction from 15 $\rightarrow$ 12 $\rightarrow$ 10 prompts per day reflects lessons learned about participant burden. Fewer prompts with longer minimum gaps improved per-prompt response quality. Total label volume was maintained by increasing participant counts.
- **Hourly upload cadence from D-2 onward.** D-1 used periodic background batch uploads, which occasionally produced multi-hour data gaps. Switching to hourly synchronisation in D-2 and D-3 improved data completeness and enabled closer real-time compliance monitoring.

- **ESM questionnaire expansion.** D-1 used an 8-item instrument. In D-2, the single post-ESM Change item was split into ChangedValence and ChangedArousal. D-3 introduced eight PANAS-style discrete affect-word items and rescaled Stress and Disturbance from $-3/ + 3$ to $0/ + 6$.

A.4 Study Intrinsic Bias

We discuss potential sources of intrinsic bias in our datasets that readers and future users should be aware of.

1. **Recruitment and Population Bias.** Participants were recruited via a campus bulletin board at a single South Korean university, representing a self-selected subset of a narrow, young adult student population. Findings may not generalise to other age groups, cultural contexts, or non-student populations.
2. **Gender Imbalance.** Male participants outnumber female participants across all three waves in both recruited and retained samples; after QC, the counts are D-1: 64M/28F, D-2: 75M/24F, and D-3: 67M/39F.
3. **Android-Only Bias.** The requirement for an Android smartphone excludes all iOS users, which may introduce bias related to platform-specific usage patterns and sensing coverage.
4. **Temporal and Contextual Bias.** All three waves were collected under distinct COVID-19 conditions and in winter months, meaning affective baselines, mobility, and social behaviour may not reflect typical everyday life and are not directly comparable across seasons.
5. **Compliance and Exclusion Bias.** Quality-control exclusions based on low ESM response rates and sensor coverage thresholds are non-random, likely removing participants with higher affective instability or greater device burden, and thus biasing the dataset toward participants with more stable behavioural patterns.

Table 10: Post-study psychological baseline questionnaires.

Scale	Full Name	Items	Scoring Range	Interpretation	Waves
K-BFI-15	Korean Big Five Inventory (15-item)	15	Per trait: 3–15	Measures five personality traits: Openness, Conscientiousness, Neuroticism, Extraversion, and Agreeableness. Each trait is scored by summing its three constituent items. Higher scores indicate a stronger tendency toward that trait.	D-1, D-2, D-3
PSS-10	Perceived Stress Scale (10-item)	10	0–40	Measures perceived stress over the past month using five-point Likert items. Scored by summing all responses. Higher scores indicate greater perceived stress. Scores ≥ 20 suggest elevated stress.	D-1, D-2, D-3
GHQ-12	General Health Questionnaire (12-item)	12	0–36	Measures general psychological distress using four-point Likert items. Scores of 11–12 are typical; scores > 15 indicate distress; scores > 20 indicate severe problems.	D-1, D-2, D-3
PHQ-9	Patient Health Questionnaire (9-item)	9	0–27	Measures depression symptom severity. Severity: 0–4 minimal; 5–9 mild; 10–14 moderate; 15–19 moderately severe; 20–27 severe.	D-1, D-2 only
CESD-R	CES-D Scale – Revised (Eaton et al., 2004)	20	0–60	Measures depression severity over the past week. Scores ≥ 16 indicate clinically significant symptoms. Replaces PHQ-9 from D-3 onward.	D-3 only
SWLS-overall	Satisfaction with Life Scale (Diener et al., 1985)	5	5–35	Global life-satisfaction measure; seven-point Likert items. Higher scores indicate greater satisfaction.	D-1 only
SWLS-present	SWLS present-focused subscale	3	3–21	Present-focused subset of SWLS items. Higher = greater satisfaction with current life.	D-1 only
SWLS-past	SWLS past-focused subscale	2	2–14	Past-focused subset of SWLS items. Higher = greater satisfaction with past life.	D-1 only
RSES	Rosenberg Self-Esteem Scale (Rosenberg, 1965)	10	10–40 (1–4 per item)	Measures global self-esteem. Thresholds: 15–25 normal; below 15 suggests low self-esteem.	D-1 only
PCQ (PPC-*)	Psychological Capital Questionnaire (Luthans et al., 2007); subscales: self-efficacy, optimism, hope, resiliency	6 each (self-efficacy, optimism, hope); 3 (resiliency)	Self-efficacy / optimism / hope: 6–30; resiliency: 3–18	Measures positive psychological resources. In D-1: stored as PPC-self_efficacy, etc. In D-3: stored without prefix (self-efficacy, etc.).	D-1, D-3 only

Note. Depression was measured with PHQ-9 in D-1/D-2 and replaced by CESD-R in D-3. SWLS (overall/present/past), RSES, and PCQ (PPC-*) are D-1 only; the four PCQ subscales reappear in D-3 without prefix.

A.5 Survey Details

This section describes all surveys administered across the three waves. Surveys fall into two categories: post-study psychological baseline questionnaires collected once per participant at the end of each wave, and ESM label instruments collected repeatedly throughout the study period.

A.5.1 Post-Study Psychological Baseline Questionnaires

The post-study psychological baseline questionnaires are summarised in Table 10 above. These were administered after the data collection period ended, and responses are stored in `UserInfo.csv` per wave.

A.5.2 ESM Label Instruments

Affective and cognitive state labels were collected via ESM prompts delivered through ABC Logger throughout each wave. The ESM questionnaire differs across waves while preserving a shared core label set to support cross-wave benchmarking.

Table 11: D-1 ESM questionnaire fields (8 items).

Field	Scale	Description
Valence	−3 to +3	Emotion valence: very negative → very positive
Arousal	−3 to +3	Emotion arousal: very calm → very excited
Stress	−3 to +3	Stress level: not stressed at all → very stressed
Duration	0–60 min / N/A	Duration of the current emotional state in minutes
Attention	−3 to +3	Attention level: very bored → very engaged
Disturbance	−3 to +3	Task disturbance: did not disturb at all → very disturbing
Mental	−3 to +3	Mental load of the ongoing task: very low → very high
Change	−3 to +3	Emotion change while answering: felt more negative → more positive

Table 12: D-2 ESM questionnaire fields (9 items).

Field	Scale	Description
Valence	−3 to +3	Emotion valence: very negative → very positive
Arousal	−3 to +3	Emotion arousal: very calm → very excited
Stress	−3 to +3	Stress level: not stressed at all → very stressed
Duration	0–60 min / N/A	Duration of the current emotional state in minutes
Attention	−3 to +3	Attention level: very bored → very engaged
Disturbance	−3 to +3	Task disturbance: did not disturb at all → very disturbing
Mental	−3 to +3	Mental load of the ongoing task: very low → very high
ChangedValence	−3 to +3	Valence state immediately after answering the ESM. Replaces the single Change item from D-1.
ChangedArousal	−3 to +3	Arousal state immediately after answering the ESM. Replaces the single Change item from D-1.

Table 13: D-3 ESM questionnaire fields (12 items).

Field	Scale	Description
<i>PANAS-style affect items (new in D-3)</i>		
Happy	0 to +6	Intensity of feeling happy
Relaxed	0 to +6	Intensity of feeling relaxed
Cheerful	0 to +6	Intensity of feeling cheerful
Content	0 to +6	Intensity of feeling content
Sad	0 to +6	Intensity of feeling sad
Anxious	0 to +6	Intensity of feeling anxious
Depressed	0 to +6	Intensity of feeling depressed
Angry	0 to +6	Intensity of feeling angry
<i>Core labels (shared with D-1 and D-2)</i>		
Valence	−3 to +3	Emotion valence: very negative → very positive. Scale unchanged from D-1/D-2.
Arousal	−3 to +3	Emotion arousal: very calm → very excited. Scale unchanged from D-1/D-2.
Stress	0 to +6*	Stress level. Scale changed from −3/ + 3 to 0/ + 6 relative to D-1/D-2.
Disturbance	0 to +6*	Task disturbance level. Scale changed from −3/ + 3 to 0/ + 6 relative to D-1/D-2.

A.6 Sensor and Modality Details

All raw sensor data are stored as per-participant UTF-8 encoded CSV files. Timestamps across all files use Unix epoch time in UTC+0 (milliseconds since 1 January 1970). Data are organised into two sources: smartphone streams collected via ABC Logger, and wearable streams from the Fitbit Inspire HR. Smartphone streams are further categorised into system sensing, interaction sensing, and context sensing.

Figure 2 (in the main text) provides a visual summary of sensor and modality availability across the three waves.

A.6.1 Smartphone Sensor Files

Table 14: System sensing files.

File	Description	Waves	Key Fields
BatteryEvent.csv	Battery level and charging state (event-based)	D-1, D-2, D-3	status (CHARGING / DISCHARGING / FULL / NOT_CHARGING), temperature (dec ⁻ °C), level (%), plugged (AC / USB / UNDEFINED)
InstalledApp.csv	Installed applications list, queried twice per day	D-1, D-2, D-3	name, packageName, isSystemApp, isUpdatedSystemApp, firstInstallTime, lastUpdateTime

Table 15: Interaction sensing files.

File	Description	Waves	Key Fields
AppUsageEvent.csv	App foreground / background state changes	D-1, D-2, D-3	name, packageName, isSystemApp, type (ACTIVITY_PAUSED, ACTIVITY_RESUMED, SCREEN_INTERACTIVE, SCREEN_NON_INTERACTIVE, ...)
NotificationEvent.csv	App notification history	D-1, D-2, D-3	name, packageName, visibility (PUBLIC / PRIVATE / SECRET / UNDEFINED), category (alarm / call / email / msg / social / ...)
CallEvent.csv	Incoming / outgoing / missed call history	D-1, D-2, D-3	type (INCOMING / OUTGOING / MISSED / UNDEFINED), number (MD5-hashed), isPinned, isStarred, duration (s), dataUsage
MessageEvent.csv	SMS / MMS history	D-1, D-2, D-3	number (MD5-hashed), messageClass (SMS / MMS / UNDEFINED), messageBox (INBOX / SENT / UNDEFINED)
DeviceEvent.csv	System-level device state changes	D-1, D-2, D-3	type (SCREEN_ON / SCREEN_OFF, BLUETOOTH_ACTIVATE / DEACTIVATE, POWER_CONNECTED / DISCONNECTED, RINGER_MODE_NORMAL / SILENT / VIBRATE, ...)
KeyEvent.csv	Keystroke touch interaction events	D-2, only D-3	keyboardType (CHUNJIIN / QWERTY_KOR / OTHERS), prevKeyType / currKeyType (BACKSPACE / KOR / ENG / NUMBER / SPECIAL), distance (key-grid), timeTaken (ms)

Table 16: Context sensing files.

File	Description	Sampling Rate	Waves	Key Fields
Light.csv	Ambient light intensity	5 Hz	D-1, D-2, D-3	brightness (lux)
ActivityEvent.csv	Physical activity type with confidence	Event-based	D-1, D-2, D-3	type (STILL / WALKING / RUNNING / IN_VEHICLE / ON_BICYCLE / TILTING / UNKNOWN), confidence (0–100%)
ActivityTransition.csv	Activity state transitions	Event-based	D-1, D-2, D-3	type, isEntered (TRUE / FALSE)
Location.csv	GPS location changes	Event-based	D-1, D-2, D-3	latitude, longitude [†] , altitude, accuracy (m), speed (m/s)
DataTraffic.csv	Network data usage	Periodic	D-1, D-2 [‡] , D-3	fromTime, toTime, rxBytes, txBytes, mobileRxBytes, mobileTxBytes
WiFiScan.csv	Nearby Wi-Fi access points	~1/min	D-1, D-2, D-3	bssid [§] (UUID-replaced), frequency (MHz), rssi (dBm)
BluetoothScan.csv	Nearby Bluetooth devices	~1/min	D-1, D-2, D-3	address [§] (UUID-replaced), rssi (dBm)

[†] GPS longitude is offset by a fixed random per-participant constant displacement to preserve relative spatial structure without exposing precise locations.

[‡] Network traffic data experienced a collection application malfunction in D-2 and should be treated with caution for that wave.

[§] Wi-Fi BSSID and Bluetooth MAC addresses are replaced with UUIDs using the Python uuid library.

A.6.2 Wearable Sensor Files

Table 17: Wearable sensor files.

File	Sensor	Sampling Rate	Waves	Description
Fitbit-HeartRate.csv	Fitbit Inspire HR	1 Hz	D-1, D-2, D-3	Heart rate (BPM)
Fitbit-StepCount.csv	Fitbit Inspire HR	1/60 Hz	D-1, D-2, D-3	Steps walked per minute
Fitbit-Calorie.csv	Fitbit Inspire HR	1/60 Hz	D-1, D-2, D-3	Calories burned per minute; activity level (0–3); METs
Fitbit-Distance.csv	Fitbit Inspire HR	1/60 Hz	D-1, D-2, D-3	Distance traveled per minute (meters)

A.6.3 Privacy Transformations

The following privacy-preserving transformations were applied to the raw data before release:

- **GPS location** – Longitude values received a fixed random per-participant constant displacement. This generates a pseudo-location trace that preserves relative distances and allows location clustering while concealing absolute geographic positions.
- **Call and message contact identifiers** – Contact numbers in `CallEvent.csv` and `MessageEvent.csv` were one-way hashed using MD5. Original digits cannot be recovered, but within-contact consistency (e.g., call frequency counts) is preserved.
- **Wi-Fi and Bluetooth hardware addresses** – Wi-Fi BSSID values and Bluetooth MAC addresses were replaced with UUIDs using the Python uuid library. This allows co-presence and proximity density estimation without exposing device identities.

A.7 Feature Extraction Details

A.7.1 Pipeline Overview

The raw sensor CSV files described in Section A.6 are processed into benchmark-ready feature matrices using the preprocessing and feature extraction pipeline described in Zhang et al. (2024). The pipeline converts per-participant raw smartphone and wearable sensor streams into fixed-format feature vectors aligned to each ESM prompt. All benchmark features are derived from sensor data available before the corresponding ESM prompt; prior ESM label values are not used as input features in the reported benchmark.

A.7.2 Feature Naming Convention

Every feature column in the benchmark-ready files follows a structured naming convention based on feature group, descriptor, optional categorical value, and optional time-window suffix:

$$\{\text{FEATURE_GROUP}\}\#\{\text{DESCRIPTOR}\}[\text{={VALUE}}][\#\{\text{WINDOW}\}]$$

In the released feature files, some historical feature names preserve implementation-specific suffixes using either # or _ before the window name. During cross-wave alignment, these names are canonicalized before constructing the shared feature intersection.

- **FEATURE_GROUP** – the feature group identifier, such as CALL_DUR, APP_CAT, or FitbitHeartrate.
- **DESCRIPTOR** – the feature descriptor. Numeric descriptors include VAL (instantaneous value), AVG (mean), STD (standard deviation), SKW (skewness), KUR (kurtosis), MED (median), BEP (binned entropy), ASC (absolute sum of changes), and TSC (time-series complexity). Categorical descriptors include one-hot value indicators and categorical summary descriptors such as entropy, duration-since-change, and relative-support features.
- **=VALUE** (optional) – appended for categorical features to specify the encoded category. For example, APP_CAT#VAL=SOCIAL indicates that the foreground app belongs to the social category.
- **WINDOW** (optional) – the time window over which the descriptor is computed. The pipeline uses two rolling windows immediately preceding the prompt, ImmediatePast_15 and ImmediatePast_30, together with six three-hour epoch windows: Dawn, Morning, Afternoon, LateAfternoon, Evening, and Night. These epoch windows are generated for both the current day and the previous day using Today and Yesterday prefixes.

A.7.3 Feature Categories, Prompt Alignment, and Time Windows

The feature groups in Table 19 are organised into six sensing categories. Each feature vector is aligned to an ESM prompt at time t , and all benchmark features are extracted only from sensor records observed at or before t . The current feature extraction pipeline uses the following active windows.

- **Immediate-past windows.** Two short look-back windows are used: ImmediatePast_15, covering $[t - 15\text{min}, t]$, and ImmediatePast_30, covering $[t - 30\text{min}, t]$. Other short windows such as 5, 10, and 45 minutes are not used in the reported benchmark.
- **Previous-day epoch windows.** Six fixed three-hour epochs from the previous calendar day are used: Dawn (06:00–09:00), Morning (09:00–12:00), Afternoon (12:00–15:00), LateAfternoon (15:00–18:00), Evening (18:00–21:00), and Night (21:00–24:00). These features use the Yesterday prefix.
- **Current-day epoch windows.** The same six three-hour epochs are generated for the current day using the Today prefix. Only epochs whose start time is no later than t are emitted, and the final active epoch is clipped at t so that no post-prompt sensor records are used.
- **Point-in-time features.** Some descriptors do not use an aggregation window. These include #VAL, the last observed value at or before t ; #DSC and #DSC={c}, the time since the last state change for categorical streams; and calendar descriptors such as Time#DOW, Time#WKD, and Time#HRN.

- **Sleep features.** Sleep-related features, including Sleep#Duration, Sleep#Onset, and Sleep#Midpoint, are derived from screen-event patterns over the overnight window from the previous day 21:00 to the current day 14:00.

Across these windows, the pipeline computes statistical and temporal descriptors including mean, standard deviation, skewness, kurtosis, median, binned entropy, absolute sum of changes, and time-series complexity, together with categorical summaries where applicable. Features derived from prior ESM labels, such as last observed label values, prior HIGH-label likelihoods, or rolling ESM-label means and standard deviations, are not included in the reported benchmark feature matrices.

Table 18: Feature categories and raw data sources.

Category	Raw Data Sources	Waves
Social interaction	CallEvent.csv, MessageEvent.csv	D-1, D-2, D-3
Physical activity	ActivityEvent.csv, ActivityTransition.csv, Fitbit-StepCount.csv, Fitbit-Distance.csv, Fitbit-Calorie.csv	D-1, D-2, D-3
Context and mobility	Location.csv, DataTraffic.csv	D-1, D-2, D-3 (DataTraffic in D-2 treated with caution – see Table 19 footnote ‡)
Phone usage	AppUsageEvent.csv, InstalledApp.csv, DeviceEvent.csv, BatteryEvent.csv, NotificationEvent.csv, KeyEvent.csv	D-1, D-2, D-3 (KeyEvent: D-2, D-3 only)
Physiological	Fitbit-HeartRate.csv	D-1, D-2, D-3
Wireless environment	WiFiScan.csv, BluetoothScan.csv	D-1, D-2, D-3 (WiFiScan: D-1, D-2 only)

A.7.4 Feature Groups and Source Files

Table 19: Mapping of feature groups to source CSV files.

CSV Source	Feature Group	Description	Waves
<i>Location.csv</i>	LOC_CLS	Location class label	D-1, D-2, D-3
	LOC_DST	Distance between consecutive location readings	D-1, D-2, D-3
	LOC_LABEL	Semantic location label (home / work / other)	D-1, D-2, D-3
<i>AppUsageEvent.csv</i>	APP_CAT	App usage category indicator	D-1, D-2, D-3
	APP_DUR_SOCIAL	App usage duration: social category	D-1, D-2, D-3
	APP_DUR_WORK	App usage duration: work category	D-1, D-2, D-3
	APP_DUR_INFO	App usage duration: information category	D-1, D-2, D-3
	APP_DUR_ENTER	App usage duration: entertainment category	D-1, D-2, D-3
	APP_DUR_HEALTH	App usage duration: health category	D-1, D-2, D-3
	APP_DUR_SYSTEM	App usage duration: system category	D-1, D-2, D-3
<i>InstalledApp.csv</i>	APP_DUR_UNKNOWN	App usage duration: unknown category	D-1, D-2, D-3
	INST_JAC	Installed apps Jaccard similarity between consecutive readings	D-1, D-2, D-3
<i>ActivityEvent.csv</i>	ACE_WLK	Activity type confidence: walking	D-1, D-2, D-3
	ACE_RUN	Activity type confidence: running	D-1, D-2, D-3
	ACE_VHC	Activity type confidence: in vehicle	D-1, D-2, D-3
	ACE_FOT	Activity type confidence: on foot	D-1, D-2, D-3
	ACE_BCC	Activity type confidence: bicycling	D-1, D-2, D-3
	ACE_TLT	Activity type confidence: tilting	D-1, D-2, D-3
	ACE_UNK	Activity type confidence: unknown	D-1, D-2, D-3
<i>ActivityTransition.csv</i>	ACT	Activity transition event type	D-1, D-2, D-3
<i>Fitbit-HeartRate.csv</i>	FitbitHeartRate	Fitbit heart rate	D-1, D-2, D-3
<i>Fitbit-StepCount.csv</i>	FitbitStepcount	Fitbit step count	D-1, D-2, D-3
<i>Fitbit-Calorie.csv</i>	Fitbitcalorie	Fitbit calories burned	D-1, D-2, D-3
<i>Fitbit-Distance.csv</i>	Fitbitdistance	Fitbit distance traveled	D-1, D-2, D-3
<i>CallEvent.csv</i>	CALL_CNT	Number of calls	D-1, D-2, D-3
	CALL_DUR	Total call duration	D-1, D-2, D-3
<i>MessageEvent.csv</i>	MSG_SNT	Sent messages	D-1, D-2, D-3
	MSG_RCV	Received messages	D-1, D-2, D-3
	MSG_ALL	Total messages (sent + received)	D-1, D-2, D-3
<i>BluetoothScan.csv</i>	BT_RSSI	Bluetooth RSSI signal strength	D-1 only
	BT_BondState	Bluetooth device bond state	D-2, D-3 only
	BT_DeviceType	Bluetooth device type	D-2, D-3 only
	BT_classType	Bluetooth device class type	D-2, D-3 only
<i>DataTraffic.csv</i>	DATA_RCV	Total data received	D-1, D-2, D-3 [‡]
	DATA_SNT	Total data sent	D-1, D-2, D-3 [‡]
	DATA_MRCV	Mobile data received	D-1, D-2, D-3 [‡]
	DATA_MSNT	Mobile data sent	D-1, D-2, D-3 [‡]
<i>WiFiScan.csv</i>	WIFI_JAC	Wi-Fi BSSID Jaccard similarity	D-1, D-2 only [†]
	WIFI_COS	Wi-Fi RSSI cosine similarity	D-1, D-2 only [†]
	WIFI_EUC	Wi-Fi RSSI Euclidean distance	D-1, D-2 only [†]
	WIFI_MAN	Wi-Fi RSSI Manhattan distance	D-1, D-2 only [†]
<i>DeviceEvent.csv</i>	SCR	Screen event count	D-1, D-2, D-3
	SCR_DUR	Screen-on usage duration	D-1, D-2, D-3
<i>NotificationEvent.csv</i>	Notification_CAT	Notification category indicator	D-1, D-2, D-3
	Notification_VIS	Notification visibility indicator	D-1, D-2, D-3
<i>KeyEvent.csv</i>	keyevent_CAT	Key event app category	D-2, D-3 only
	keyevent_DIST	Keystroke distance	D-2, D-3 only
	keyevent_TIME	Keystroke inter-key time delta (ms)	D-2, D-3 only
<i>BatteryEvent.csv</i>	BAT_LEV	Battery level (%)	D-1, D-2, D-3
	BAT_TMP	Battery temperature (°C)	D-1, D-2, D-3
	BAT_PLG	Battery plug state	D-1, D-2, D-3
	BAT_STA	Battery status	D-1, D-2, D-3

[‡] D-2 DataTraffic is unreliable due to a logger malfunction.

[†] D-3 Wi-Fi logs use a different schema; Wi-Fi similarity features are D-1/D-2 only.

A.7.5 Feature Counts Per Wave

Table 20: Feature counts per wave.

Dataset	Total feature columns	Columns shared across all three waves
D-1	8,037	4,890
D-2	10,122	4,890
D-3	10,581	4,890

Differences in feature counts across waves arise from sensor availability differences and minor logger-format differences across study waves. In particular, KeyEvent-derived features are available only from D-2 onward, Wi-Fi similarity features are available in D-1/D-2, Bluetooth feature groups differ between D-1 and later waves, and minor changes to the data collection application affected the set of collectible device-event types. For cross-wave evaluation, these wave-specific feature spaces are canonicalized and intersected as described in Appendix C.1.1.

A.7.6 Label Binarisation

All ESM labels are first mapped to a common numeric scale before binarisation. Valence and arousal are already collected on a -3 to $+3$ scale across all three waves. For D-3, Stress, Disturbance, and the PANAS-style affective intensity items (Happy, Relaxed, Cheerful, Content, Sad, Anxious, Depressed, and Angry) were originally collected on a 0 to $+6$ scale; these labels are linearly rescaled to -3 to $+3$ before thresholding. This harmonization ensures that the binary targets are defined on comparable label encodings across waves.

We construct binary benchmark targets using a user-specific mean split. For each participant and label, responses above that participant’s mean for that label are encoded as HIGH (1), and all remaining responses are encoded as LOW (0). Thus, thresholds are user- and label-specific, and the resulting target represents whether a momentary affect report is above or below that participant’s own typical level.

Binarisation thresholds:

- **Temporal benchmark.** Because the temporal setting splits each user’s samples chronologically into train/validation/test partitions, the user-specific threshold is computed from that user’s training slice only. The same training-derived threshold is then applied to that user’s validation and test samples: HIGH (1) if value $>$ threshold, and LOW (0) otherwise.
- **Within-wave and cross-wave benchmarks.** Because users are disjoint between training and test folds in the within-wave benchmark and between source and target waves in the cross-wave benchmark, we compute the user-specific mean separately for each participant and use it to define that participant’s relative HIGH/LOW labels. This step defines the prediction target within each participant; it is not used as an input feature and does not create user-overlap leakage between training and test samples.

The resulting binary label columns are used only as prediction targets. They are not used to construct input features in the reported benchmark.

A.7.7 Missing Sensor Data Handling

Missingness arises naturally in passive sensing because streams are event-based, devices may be offline, permissions may be unavailable, and wearable or smartphone uploads may be interrupted. The feature extraction pipeline handles missingness at the stream, window, and feature-matrix levels.

First, raw streams are resampled at sensor-specific rates before feature computation. Duplicate timestamps are resolved by keeping the last observation. Numeric streams are resampled by averaging within each interval, linearly interpolated across gaps, and then stripped of leading or trailing missing values. Categorical streams are resampled using forward fill. If invalid values such as remaining NaN or infinite values are encountered during descriptor extraction, the corresponding window-level feature is treated as missing.

Second, point-in-time features use fixed fallback values when no valid prior observation is available. For #VAL, the fallback is 0. For duration-since-change descriptors, #DSC uses a sentinel value of -1 when no prior state change can be identified.

Third, after per-window feature extraction, remaining invalid entries are sanitized before constructing the benchmark-ready matrix. Numeric missing or invalid values are set to 0.0, while object or Boolean missing values are set to False. If an entire sensor stream is unavailable for a participant, features depending on that stream are absent for that participant and are filled during the final feature-matrix alignment. These fixed missing-value rules are applied uniformly and do not use ESM label values.

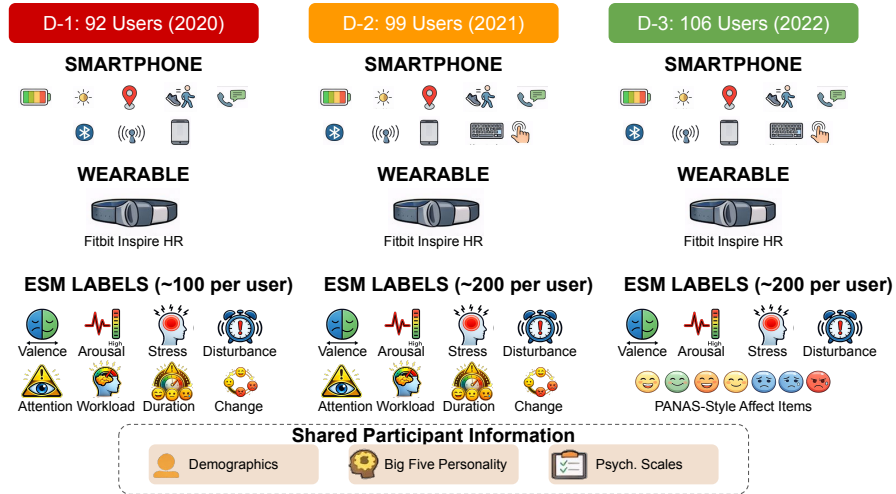


Figure 3: Overview of the EmoPhone study design across three collection waves. Each wave combines smartphone sensing, wearable sensing, and ESM affect labels, with shared participant-level information collected across waves. D-3 additionally includes PANAS-style affect descriptors.

B Additional Dataset Characterization

This appendix provides extended empirical characterization of the three-wave dataset across six dimensions: dataset overview, label richness, data density and participation, sensor coverage, distributional shift across waves, and individual variability. Each subsection is accompanied by representative figures from the full exploratory data analysis.

B.1 The Multi-Year Dataset: Overview

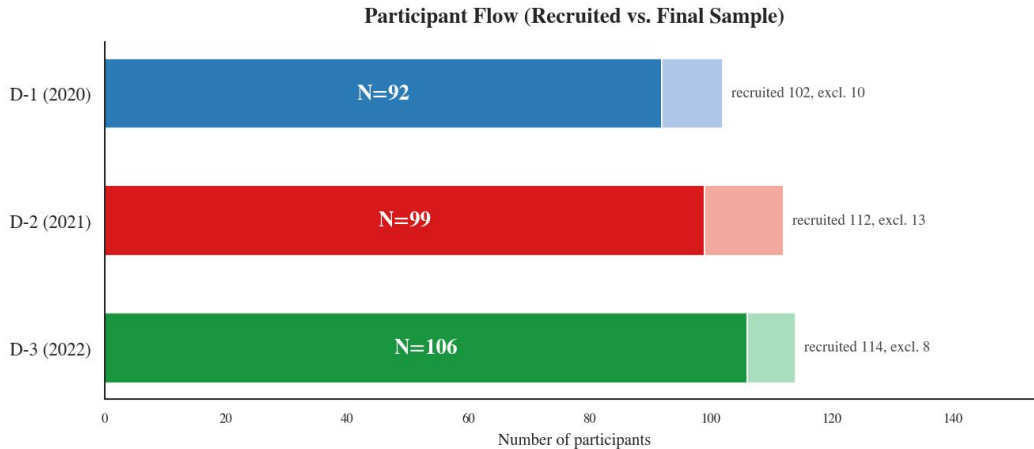


Figure 4: Participant flow from recruitment to final analysis-eligible sample per wave. Solid bars show the final retained sample (D-1: $N = 92$, D-2: $N = 99$, D-3: $N = 106$); lighter extensions show excluded participants. Exclusion rates were 9.8%, 11.6%, and 7.0% for D-1, D-2, and D-3 respectively.

Figure 3 demonstrates the study design protocol. Figure 4 shows the participant flow from recruitment to the final analysis-eligible sample across all three collection waves. A total of 328 participants were recruited (D-1: 102, D-2: 112, D-3: 114), of whom 297 remained after quality-control (QC) filtering (D-1: $N = 92$, D-2: $N = 99$, D-3: $N = 106$). QC excluded participants with wearable data coverage below 50%, smartphone data coverage below 50%, or an ESM response rate below 30%.

Exclusion rates were modest and broadly comparable across waves (D-1: 9.8%, D-2: 11.6%, D-3: 7.0%), indicating stable recruitment and data collection procedures.

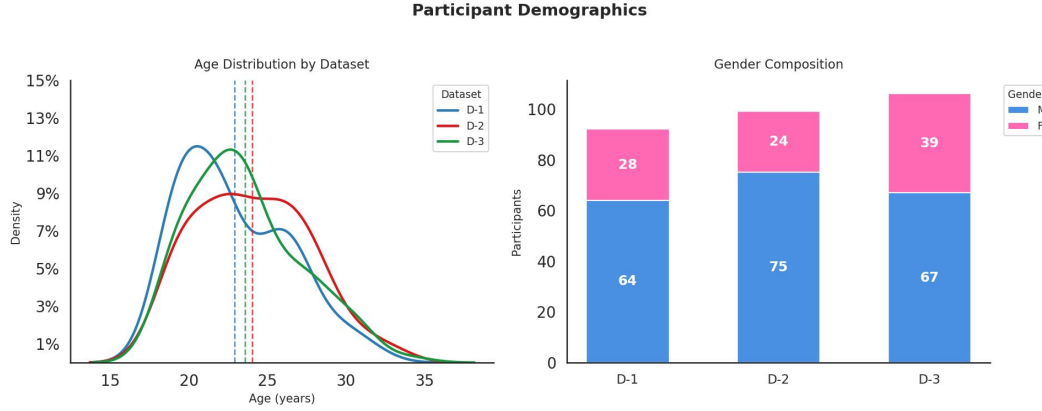


Figure 5: Participant demographics across waves. Left: age distribution per wave (line plot, dashed lines indicate wave means). Right: gender composition per wave (stacked bar). Age distributions are closely matched across waves; the female proportion ranges from 24% (D-2) to 37% (D-3).

Figure 5 characterises the age and gender composition of the retained samples. All three cohorts were drawn from the same university student population, resulting in highly comparable age distributions across waves (means: D-1 = 22.9 yr, D-2 = 24.0 yr, D-3 = 23.6 yr). The female proportion ranged from 24% to 37% across waves, with male participants constituting the majority in all three. The age-by-gender histogram confirms that the gender imbalance is consistent across age bins, rather than being concentrated in any specific age subgroup, suggesting that the imbalance reflects the broader gender composition of the source population rather than differential dropout or self-selection by gender and age jointly.

B.2 Label Richness Across Waves

The dataset employs a wave-specific ESM label instrument with a shared core across the three waves, with a shared core of four affect dimensions – Valence, Arousal, Stress, and Task Disturbance – present in all waves and constituting the primary benchmark targets. Figure 6 presents the pairwise Spearman rank correlation matrices among these four shared labels computed separately for each wave.

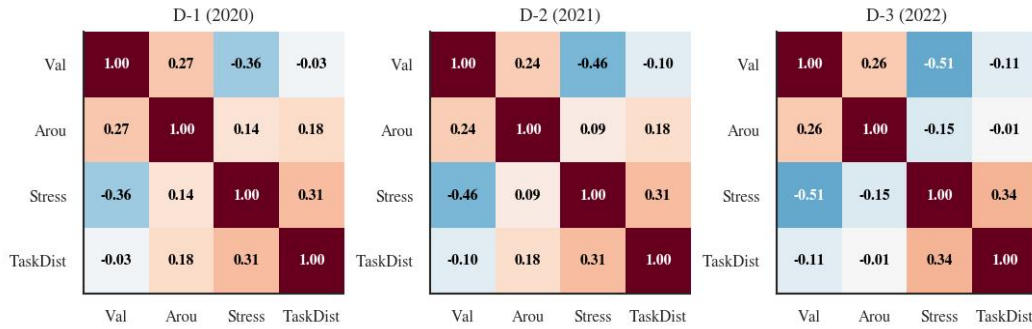


Figure 6: Pairwise Spearman rank correlation matrices for the four shared ESM labels (Valence, Arousal, Stress, Task Disturbance) computed per wave. Red cells indicate positive correlation; blue cells indicate negative correlation. The negative Valence–Stress correlation strengthens monotonically from D-1 to D-3 (ρ : $-0.35 \rightarrow -0.45 \rightarrow -0.51$).

The inter-label structure is broadly consistent across waves and reflects well-established affective relationships. Valence and Stress exhibit a moderate-to-strong negative correlation in all three waves (D-1: $\rho = -0.35$; D-2: $\rho = -0.45$; D-3: $\rho = -0.51$), with the association strengthening incrementally across collection years, consistent with the expanded PANAS-style labelling in D-3 providing finer affective resolution. Stress and Task Disturbance are positively correlated across all waves ($\rho = 0.31, 0.31, 0.35$), reflecting their conceptual proximity as negative cognitive-load states. Valence and Arousal are weakly positively correlated ($\rho = 0.24$ – 0.28), consistent with their orthogonality in the circumplex model of affect. Notably, the Arousal–Stress correlation shifts from weak positive in D-1 ($\rho = 0.15$) to near-zero or slightly negative in D-2 and D-3 ($\rho = 0.08, -0.15$), suggesting that the rescaling of the stress item in D-3 (from $-3/+3$ to $0/+6$, then back-normalised) and the addition of discrete affect words may have shifted how participants anchor their arousal responses.

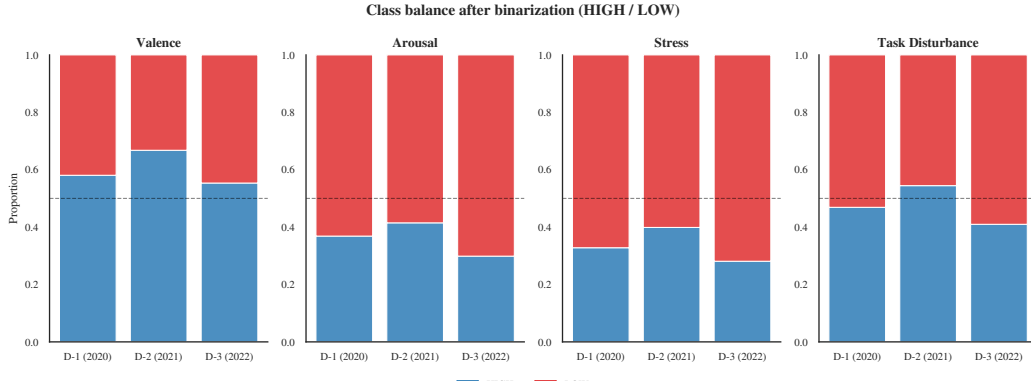


Figure 7: Class balance after user-level mean binarization for the four shared labels (Valence, Arousal, Stress, Task Disturbance) across D-1, D-2, and D-3. Each bar shows the proportion of HIGH and LOW responses per wave.

Figure 7 complements the inter-label correlation view by showing the class balance resulting from user-level mean binarization applied to all benchmark targets. For each of the four shared labels, the proportion of HIGH and LOW responses is shown separately per wave. The resulting class distributions remain reasonably balanced across labels and waves, supporting AUROC-based evaluation while preserving the interpretation of HIGH/LOW as deviations above or below each participant’s typical affective level.

B.3 Data Density and Participation

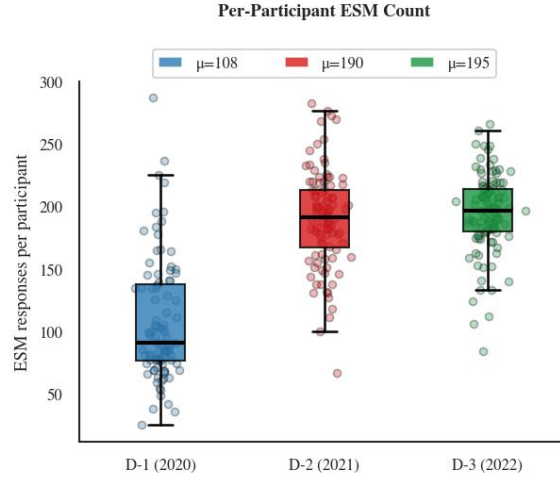


Figure 8: Per-participant total ESM response count distributions per wave (D-1: $\mu = 108$, $SD = 49$; D-2: $\mu = 190$, $SD = 39$; D-3: $\mu = 195$, $SD = 32$). Box shows IQR; individual points show per-participant values. D-2 and D-3 exhibit substantially higher and more consistent label volumes than D-1, reflecting wave-specific differences in prompt frequency and participation patterns.

Figure 8 shows the distribution of per-participant total ESM response counts across all three waves. The substantially wider and lower-centred distribution for D-1 (mean $\mu = 112$, $SD = 51$) compared to D-2 ($\mu = 213$, $SD = 44$) and D-3 ($\mu = 206$, $SD = 25$) reflects the higher ESM prompt frequency in D-1 (15/day) paired with lower per-prompt compliance, alongside the shorter effective participation windows of several D-1 participants. D-2 and D-3 show tight, well-concentrated response count distributions, with most participants contributing between 150 and 260 labelled ESM events. The marked reduction in variance from D-1 to D-3 reflects the combined effect of the participant-facing progress counter introduced in D-2, the intentional reduction in daily prompt frequency, and improved compliance monitoring protocols. Together, the D-2 and D-3 distributions confirm that the dataset provides dense, individual-level label sequences suitable for within-person temporal modelling.

B.4 Sensor Data Characterization

Figure 9 presents the cross-wave sensor and label modality coverage as a combined two-panel matrix. Panel A enumerates all sensor streams collected by the ABC Logger smartphone application and the wearable devices, indicating their presence or absence per wave. Panel B provides the corresponding label coverage matrix, identifying which ESM label dimensions were administered in each wave.

A	D-1 (2020)	D-2 (2021)	D-3 (2022)
Fitbit-HeartRate	✓	✓	✓
Fitbit-StepCount	✓	✓	✓
Fitbit-Calorie	✓	✓	✓
Fitbit-Distance	✓	✓	✓
GPS / Location	✓	✓	✓
Ambient Light	✓	✓	✓
Activity/Transition	✓	✓	✓
BluetoothScan	✓	✓	✓
WiFiScan	✓	✓	✓
KeyEvent	✗	✓	✓
Network Traffic	✓	✓	✓
Call / Message	✓	✓	✓

B	D-1 (2020)	D-2 (2021)	D-3 (2022)
Valence	✓	✓	✓
Arousal	✓	✓	✓
Stress	✓	✓	✓
Task Disturbance	✓	✓	✓
Attention	✓	✓	
Mental	✓	✓	
Duration	✓	✓	
Change	✓		
Changed Valence		✓	
Changed Arousal		✓	
Happy			✓
Relaxed			✓
Cheerful			✓
Content			✓
Sad			✓
Angry			✓
Anxious			✓
Depressed			✓

Figure 9: Sensor and label modality coverage matrices. Panel A: sensor stream availability per wave. The KeyEvent stream was added from D-2 onwards. Panel B: ESM label coverage per wave. The four shared core labels (Valence, Arousal, Stress, Task Disturbance) are available in all three waves; wave-specific labels are shown in the stepped staircase pattern.

Panel A confirms that the core smartphone sensing suite – comprising GPS/Location, Bluetooth and WiFi scans, app usage, screen events, notifications, battery events, accelerometer, ambient light, activity transitions, network traffic, and call/message metadata – was active across all three waves. Keystroke sensing (KeyEvent) was absent in D-1 and introduced from D-2 onwards. The Fitbit wrist wearable streams (heart rate, step count, calorie, distance) were available continuously throughout all three waves.

Panel B illustrates the cross-wave differences in the ESM label schema across waves. The four core dimensions (Valence, Arousal, Stress, Task Disturbance) are shared across D-1, D-2, and D-3. Attention, Mental effort, Activity Duration, and a single Change item were present in D-1 and D-2; the Change item was split into ChangedValence and ChangedArousal in D-2. D-3 introduced eight discrete PANAS-style affect-word items (Happy, Relaxed, Cheerful, Content, Sad, Angry, Anxious, Depressed), substantially enriching the affective granularity of that wave. This stepped structure means that the four shared core labels constitute the recommended benchmark targets for cross-wave evaluation, while wave-specific labels enable single-wave analyses with richer affective resolution.

B.5 Cross-Wave Differences

Across D-1, D-2, and D-3, these differences arise from participant composition, wave-specific study design choices, sensing coverage, and label design, making cross-wave generalization a meaningful setting rather than an artificial split. While they share enough common structure to support unified evaluation, they also differ in ways that make cross-wave evaluation nontrivial. Some of these differences follow directly from wave-specific study design choices: D-1 establishes the base protocol, D-2 refines questionnaire structure and increases label volume, and D-3 expands the semantic label layer. Figure 10 provides a complementary empirical view in feature space. For this visualization, each point is a participant-level aggregate representation obtained by averaging that participant’s preprocessed sensor features over the available observation period. The resulting embeddings show visible wave-associated structure rather than a single undifferentiated cloud, indicating that cross-wave differences are reflected not only in protocol design but also in the observed feature space. At the same time, the overlap between waves suggests realistic structured heterogeneity rather than trivial separability.

This subsection also analyzes cross-wave distributional shifts through three complementary lenses: covariate, concept, and conditional shift.

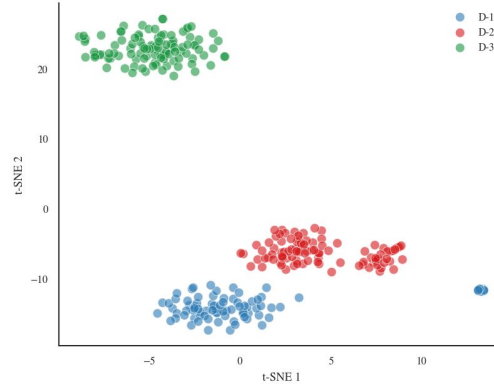


Figure 10: t-SNE visualization of participant-level sensor representations across the three waves. Each point represents one participant, obtained by averaging that participant’s preprocessed sensor features over the available observation period.

Figure 11 shows covariate shift – $P(X)$ – by overlaying the distributions of representative sensor features across the three waves. Four features spanning distinct sensing categories are shown: sleep duration (wearable/activity), total calories burned (physical activity), battery temperature (phone system), and total sent bytes (data traffic). Battery temperature and calorie distributions are closely aligned across waves, indicating stable physiological and device characteristics. In contrast, sleep duration and data traffic show pronounced wave-level shifts: D-1 participants exhibit a broader, flatter sleep distribution compared to D-2 and D-3, and data traffic volumes in D-1 are substantially higher and more variable, likely reflecting differences in COVID-19 restriction phases and associated changes in mobility and online activity patterns across the three collection years.

Figure 12 illustrates concept shift – $P(Y | X)$ – using sleep duration as the focal feature. For each of the four shared affect labels, participants are binned into low, medium, and high sleep duration groups, and the proportion reporting a HIGH affect label is plotted per bin and per wave. If the feature-label relationship were stable across waves, the three wave-level curves would overlap. Instead, the curves diverge substantially: for Valence, D-2 participants in every sleep duration bin report HIGH valence at rates 20–25 percentage points above D-3; for Arousal and Stress, D-3 participants consistently show lower HIGH rates regardless of sleep duration bin. This demonstrates that the mapping from sleep behaviour to affective state shifts across waves, meaning that a model trained on one wave cannot rely on the same feature-label associations in another.

Covariate Shift - Sensor Feature Distributions Across Waves

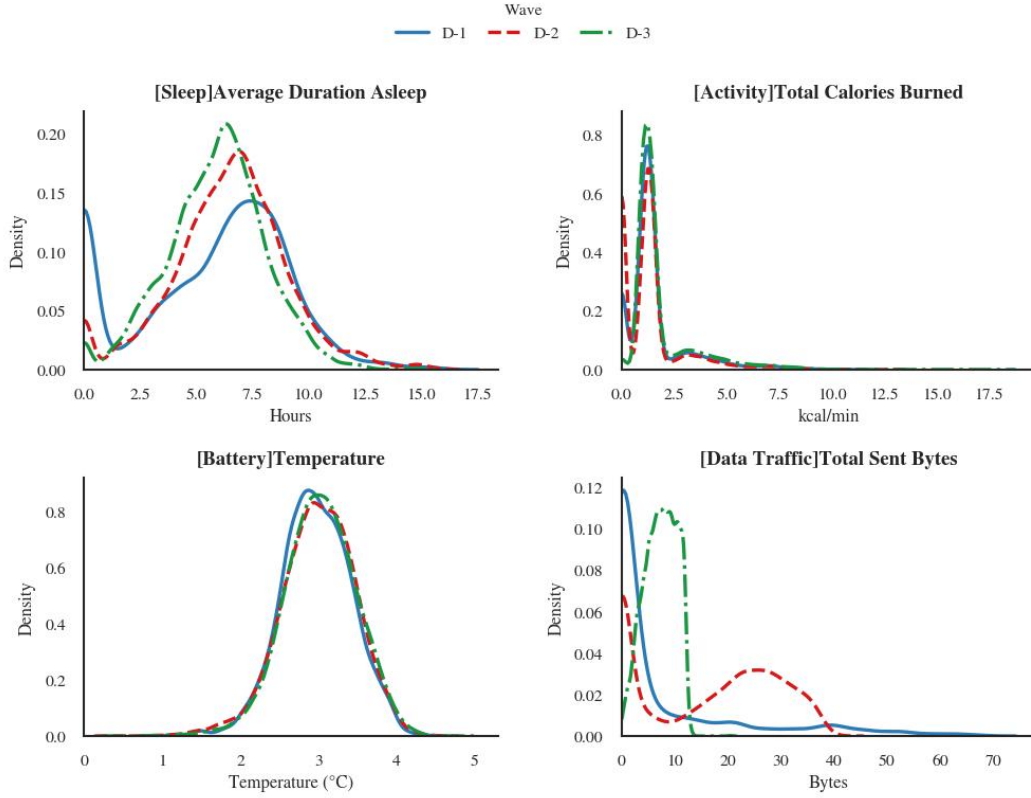


Figure 11: Covariate shift $P(X)$: kernel density estimates of four representative sensor features across the three waves (D-1 blue solid, D-2 red dashed, D-3 green dash-dot). Battery temperature and calorie features are stable across waves; sleep duration and data traffic exhibit clear wave-level distributional differences.

Concept Shift - Sleep Duration vs. High Affect Response Rate Across Waves

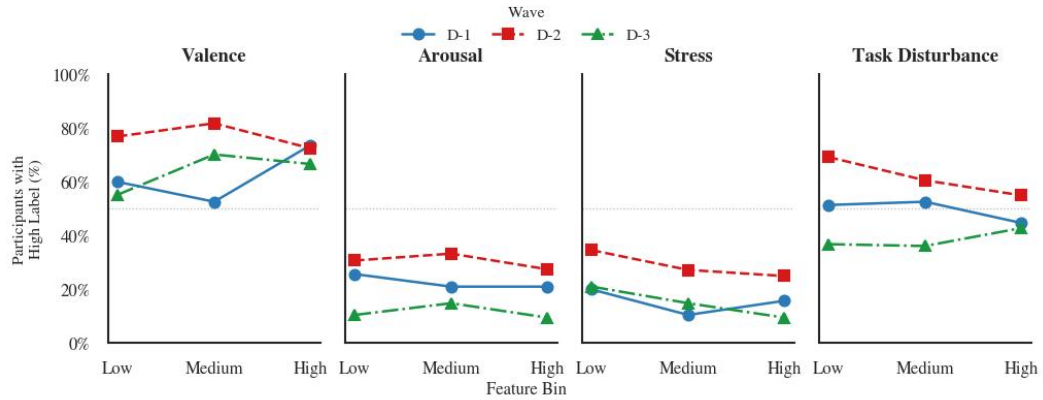


Figure 12: Concept shift $P(Y | X)$: proportion of participants reporting HIGH label as a function of binned sleep duration, shown separately for Valence, Arousal, Stress, and Task Disturbance. Each line is one wave. Substantial wave-level divergence indicates that the feature-to-label mapping is not stable across collection years.

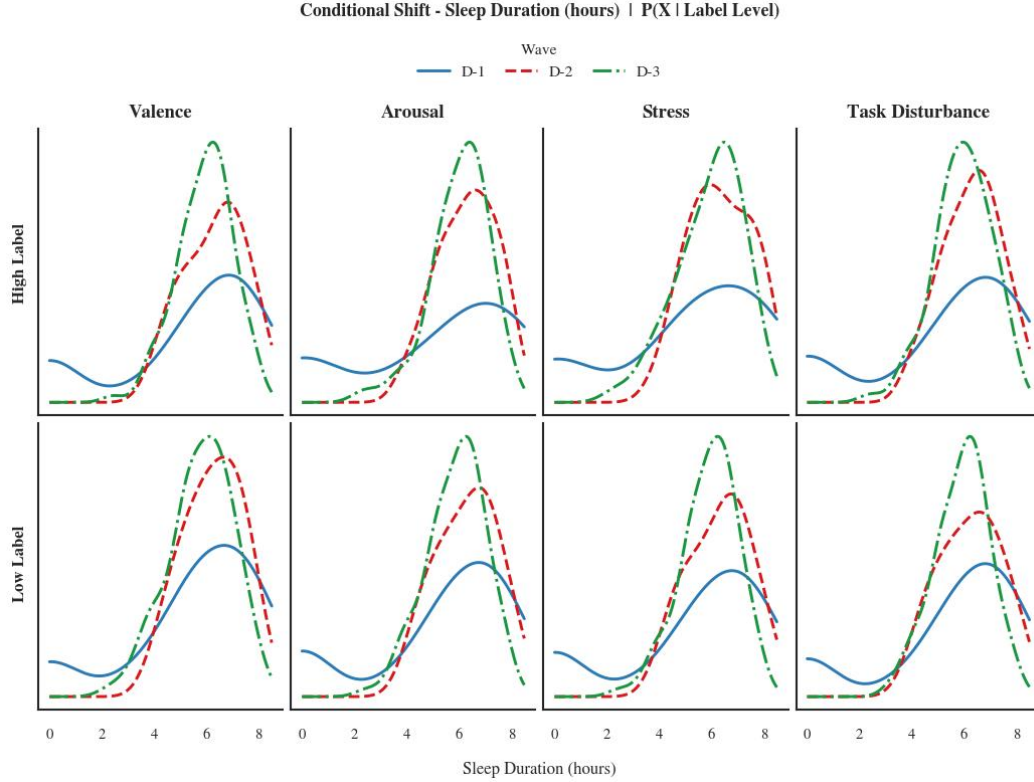


Figure 13: Conditional shift $P(X | Y)$: distribution of sleep duration conditioned on label level (HIGH top row, LOW bottom row) for each of the four shared labels, across three waves. Wave-level offsets in both rows demonstrate that sensor distributions shift across collection years independently of label level, violating the $P(X | Y)$ stability assumption required for cross-wave generalization.

Figure 13 presents conditional shift – $P(X | Y)$ – using sleep duration as the focal feature, showing how its distribution differs between HIGH and LOW label groups within each wave. The 2×4 grid of kernel density estimates (rows: HIGH vs. LOW label; columns: Valence, Arousal, Stress, Task Disturbance) reveals two distinct patterns. First, within a given wave and label level, D-2 and D-3 participants show sharper, more concentrated sleep distributions than D-1 participants, whose distributions are flatter and broader. Second, within each wave, the HIGH and LOW label groups show only modest distributional separation, consistent with sleep duration being a weak-to-moderate predictor of momentary affect. Crucially, the wave-level offset in both the HIGH and LOW rows demonstrates that conditional shift operates independently of label level: participants from different waves exhibit different sensor distributions even when conditioned on the same affective state, a pattern that directly challenges the assumption of $P(X | Y)$ stability required for naive cross-wave generalization.

B.6 Cross-User Differences Within Each Wave

Figure 14 provides an exploratory view of participant-level variability within each wave. It uses the same participant-level aggregate representation as Figure 10, but computes and visualizes the embeddings separately for D-1, D-2, and D-3. Within each wave, participants are spread from more central to more peripheral regions of the feature space rather than collapsing into a compact cluster. This pattern is visible across all three waves, indicating that within-wave variability is broad rather than being driven only by a few outliers.

These results show that substantial individual variability remains even within a relatively controlled cohort. The resource is therefore useful not only for descriptive analysis, but also for studying how affect models behave under realistic participant-level heterogeneity.

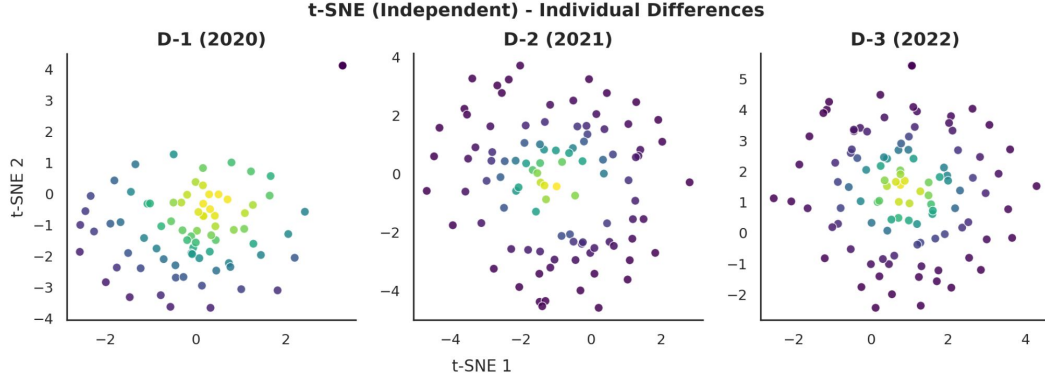


Figure 14: t-SNE visualization of participant-level sensor representations within each wave. Colors encode the participant’s distance to the wave-specific centroid in the original feature space, with brighter colors indicating more central participants and darker colors indicating more peripheral ones

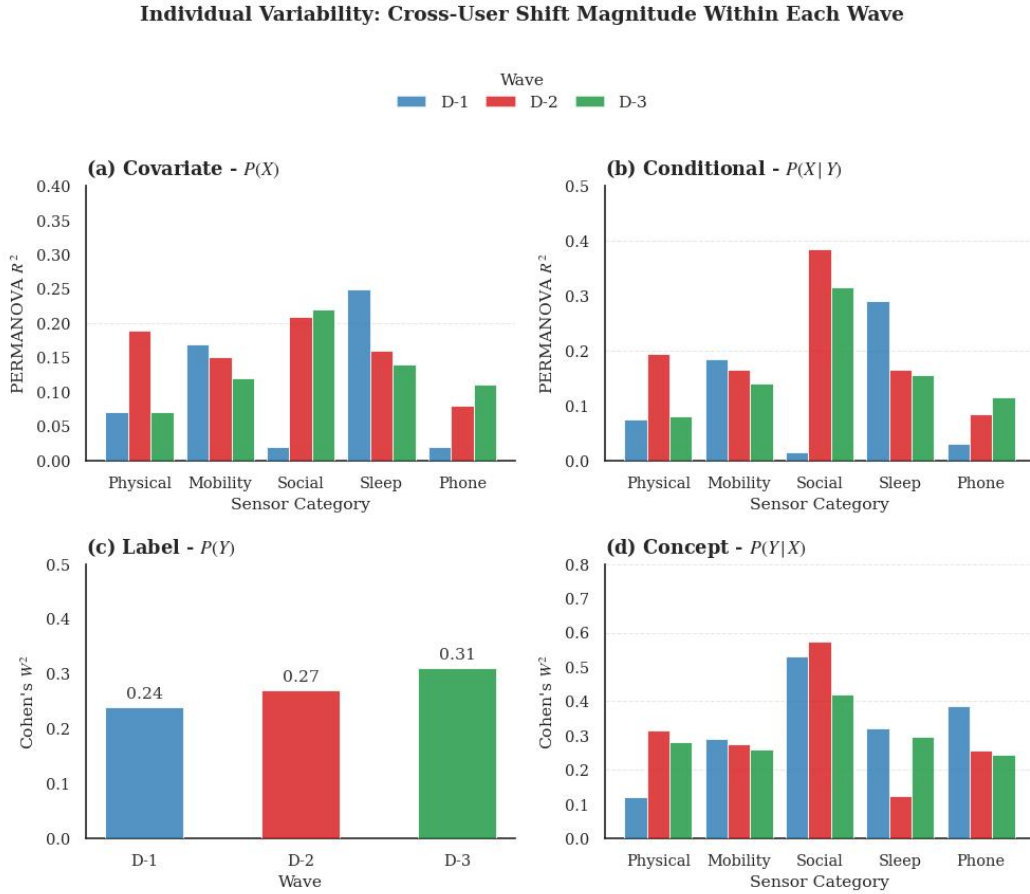


Figure 15: Cross-user shift magnitude within each wave, decomposed by shift type and sensor category. (a) Covariate shift $P(X)$: PERMANOVA R^2 of sensor feature distributions explained by participant identity. (b) Conditional shift $P(X|Y)$: PERMANOVA R^2 of conditional feature distributions. (c) Label shift $P(Y)$: Cohen’s W^2 of label distributions per wave (0.24 / 0.27 / 0.31 for D-1 / D-2 / D-3). (d) Concept shift $P(Y|X)$: Cohen’s W^2 of feature-to-label mapping variability across individuals. Social features exhibit the highest concept shift; Sleep features exhibit the highest covariate shift.

Figure 15 quantifies the magnitude of cross-user distributional shift within each wave using a four-panel analysis of shift type and sensor category. Panels (a) and (b) report PERMANOVA R^2 values measuring the fraction of variance in sensor feature distributions $P(X)$ and conditional sensor distributions $P(X | Y)$, respectively, attributable to participant identity. Panel (c) reports Cohen’s W^2 for the label distribution $P(Y)$ per wave. Panel (d) reports Cohen’s W^2 for the concept distribution $P(Y | X)$, capturing how strongly the feature-to-label mapping varies across individuals.

Across all four shift types and sensor categories, participant identity accounts for a non-trivial and consistent fraction of total variance. For covariate shift $P(X)$, sleep-related features exhibit the highest cross-user R^2 (up to 0.25 in D-1), followed by Social interaction and Mobility features. The Social category shows the strongest individual-level concept shift $P(Y | X)$ (Cohen’s $W^2 \approx 0.53$ – 0.58 across waves), indicating that social behavioural patterns have the most idiosyncratic relationship with affective states. Label shift $P(Y)$ is also substantial and increases monotonically across waves (Cohen’s $W^2 = 0.24, 0.27, 0.31$ for D-1, D-2, D-3 respectively), confirming that individuals differ systematically in their affective baselines and that this heterogeneity grows with the enrichment of the label schema. Taken together, these results establish that user identity is a dominant source of variance at every level of the feature-label pipeline, providing strong quantitative justification for personalised modelling approaches.

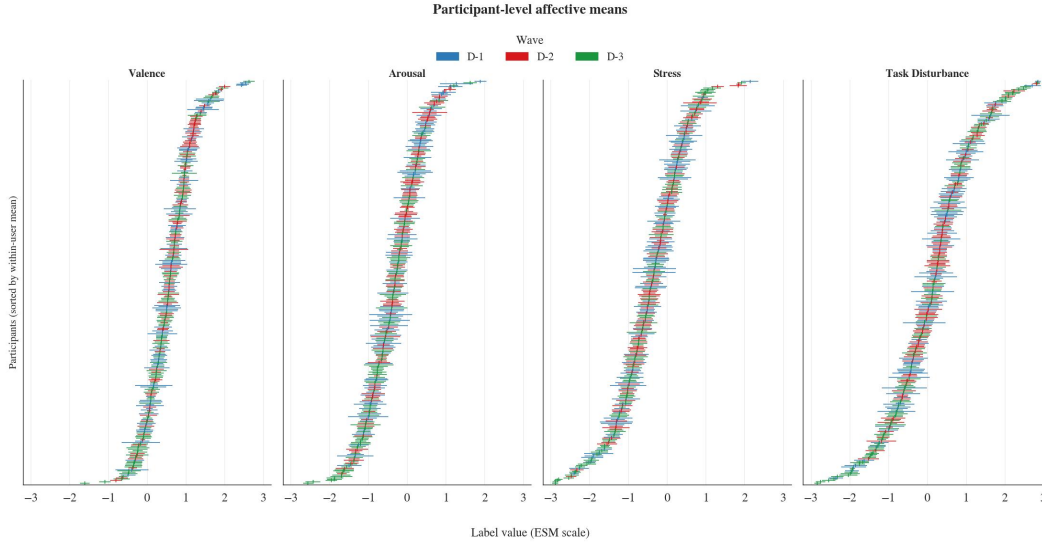


Figure 16: Participant-level affective means for the four shared labels (Valence, Arousal, Stress, Task Disturbance) across all three waves. Each horizontal bar represents one participant’s within-user mean $\pm$ one standard deviation, sorted by within-user mean value.

Figure 16 provides a complementary participant-level view of this heterogeneity, focusing directly on the label space rather than the feature space. For each of the four shared labels, each participant is represented by their within-user mean and within-user standard deviation, with participants sorted by their mean value. The wide vertical spread across all four panels confirms that individuals differ substantially in their affective baselines. Critically, the interleaving of wave colors throughout the sorted distribution indicates that individual-level spread is not simply a wave-level artifact – participants from D-1, D-2, and D-3 appear across the full range of affective means, consistent with the quantitative shift analysis in Figure 15 showing that user identity accounts for a substantial fraction of label variance within each wave.

C Benchmark Details and Full Results

C.1 Additional Benchmark Evaluation Protocol

C.1.1 Feature Alignment and Preprocessing

To ensure evaluation integrity, all preprocessing statistics are fit on the training split only and then applied unchanged to validation and test.

For the **cross-wave benchmark**, we additionally align feature spaces across waves.

Shared feature space. Raw feature sets vary by wave (Appendix A.7.5, Table 20) due to (i) sensor-availability differences (e.g., KeyEvent-derived features from D-2 onward), (ii) logger-format drift that encodes the same underlying feature under slightly different column names, and (iii) minor case/naming inconsistencies. We construct a shared cross-wave feature space with two steps.

- **Alias canonicalization.** We map each raw feature name to a canonical form by normalizing case and tokens (e.g., PIF#AGE vs. PIF#age), resolving a small set of categorical aliases (e.g., UNKNOWN vs. UNDEFINED), and removing logger artefacts. Leakage-prone columns (timestamps, participant identifiers, labels) are removed before canonicalization.
- **Feature intersection.** We use a fixed three-way intersection of canonicalized feature sets, $\mathcal{F} = \mathcal{F}_{D-1} \cap \mathcal{F}_{D-2} \cap \mathcal{F}_{D-3}$, yielding a common feature space of 4,890 features. This triple intersection is used for every cross-wave direction (both one-to-one and two-to-one). Pairwise intersections are recorded for auditing but are not used for training or evaluation.

C.1.2 Metrics and Model Selection Strategy

Evaluation metrics. We use **AUROC** as the primary metric across settings for robustness to class imbalance.

Validation and model selection. Hyperparameter tuning and checkpoint selection use only setting-defined validation data.

- **Within-wave benchmark:** We use stratified group 5-fold cross-validation, stratifying by the binary label while grouping by user identity so that label balance is preserved across folds and each user appears in exactly one fold.
- **Cross-wave benchmark:** In the two-to-one setting, we align and concatenate the two source waves into a single source pool, then hold out a stratified 20% source-only validation slice. The target wave is used only for final evaluation and does not contribute any selection signal.

C.1.3 Normalization and outlier handling We apply the same high-level preprocessing steps across settings, with setting-specific normalization. All normalization statistics are fitted only on the training split defined by each benchmark setting and are then applied unchanged to the corresponding validation and test splits. No validation/test feature statistics or affect labels are used when fitting normalization parameters.

- **Temporal benchmark normalization:** For each user, z-score statistics are computed from that user’s temporal training slice and then applied to that user’s validation and test slices.
- **Within-wave/Cross-wave benchmark normalization:** We apply per-user z-score normalization using each user’s sensor-feature statistics only, without using label values. Because users are disjoint across users in the within-wave benchmark and across waves in the cross-wave benchmark, per-user normalization does not introduce data leakage. This normalization should be interpreted as an unlabeled feature-level calibration step for retrospective or calibrated mobile-sensing evaluation, not as a fully cold-start deployment setting.
- **Outlier clipping:** We clip feature values using a training-fitted threshold based on the 99.9% quantile of $|z|$, with a minimum clipping threshold of 10, applied to all splits. This prevents a small number of extreme feature values from dominating optimization (especially for neural models) while keeping the transformation monotone and training-only to avoid leakage.

C.2 Models and training setup

C.2.1 Model implementations and architecture details

This subsection summarizes model implementations and how domain labels are instantiated. DG methods follow DomainBed-style protocols [11]. In the within-wave benchmark, each participant is treated as a domain. In the cross-wave benchmark, the train/test split is defined by wave, but DG domains are still defined at the participant level: each source participant is assigned a unique domain label, identified by its source wave and user ID. This means that both single-source and multi-source cross-wave training contain multiple source domains, corresponding to the users available in the source wave(s). UDA methods are implemented using the **Transfer-Learning-Library (TLL)** [18] under *transductive* unsupervised domain adaptation (labeled source, unlabeled target). In the within-wave benchmark, the unlabeled target set is the held-out user fold; in the cross-wave benchmark, it is the target wave. Target labels are never used for training or selection. All UDA methods receive the same unlabeled target set under the same backbone and training loop, so differences reflect the adaptation objective rather than privileged access to target information. Baselines and other tabular models are integrated from reference libraries or official repositories and wrapped in a unified pipeline.

The complete model inventory is provided in Table 21, and the hyperparameter search spaces are detailed in Tables 22–24.

C.2.1.1 Baseline models

XGBoost [5]: XGBoost is the main gradient-boosted tree baseline in the benchmark. It is included because boosted trees remain a strong reference point for dense tabular sensing features with heterogeneous scales and nonlinear interactions.

LightGBM [22]: LightGBM is a histogram-based gradient-boosted tree method that complements XGBoost with a second strong tree baseline. It is included because fast boosted-tree methods are often competitive even against more specialized tabular neural models.

MLP: We use a standard multilayer perceptron as the simplest neural baseline for fixed tabular inputs. It provides the reference feedforward model against which the tabular neural, DG, and UDA methods can be compared.

ResNet [12]: ResNet is the residual-feedforward neural baseline. It tests whether residual blocks improve optimization and performance over a plain MLP on the same feature tables.

C.2.1.2 Tabular neural networks

TabNet [2]: TabNet is an attentive tabular architecture that learns feature-selection masks over multiple decision steps. It is included as a tabular model that performs adaptive feature selection rather than uniform full-table processing.

SAINT [46]: SAINT is a transformer-style tabular model with richer attention structure than a plain feedforward network. It is included as a higher-capacity tabular transformer baseline.

TabTransformer [15]: TabTransformer contextualizes tabular features through transformer layers applied to embedded inputs. It is included because transformer-style contextualization can capture cross-feature dependencies that are difficult to model with a plain MLP.

FTTransformer [10]: FTTransformer is a lighter feature-token transformer for tabular data. It provides a strong transformer baseline with a simpler tokenization scheme than heavier row-column attention variants.

DCN [51]: DCN combines explicit cross layers with a deep nonlinear tower. It is included because the cross network can model low-order feature interactions directly.

C.2.1.3 Domain generalization models

Current benchmark runs evaluate DG models with a shared MLP backbone so that differences primarily reflect the DG objective rather than backbone changes.

Invariant Risk Minimization (IRM) [3]: IRM encourages the learned representation to support a stable classifier across source domains. It serves as the canonical invariance-based DG baseline.

Risk Extrapolation (VREx) [23]: VREx penalizes variance in risk across source domains. It is included as a close alternative to IRM based on risk equalization rather than invariant-classifier constraints.

GroupDRO [40]: GroupDRO optimizes the worst-group loss during training, treating domains as groups. It serves as the robust optimization baseline for heterogeneous source domains.

MixStyle [56]: MixStyle perturbs and mixes feature statistics across training domains. It is included as a feature-statistic augmentation baseline for DG.

MLDG [24]: MLDG is a meta-learning approach to domain generalization. It simulates train-domain and meta-test-domain splits during optimization so that updates favor held-out-domain performance.

MASF [7]: MASF combines task generalization with feature-space alignment and class-level metric learning. It is included as a stronger DG baseline that explicitly targets transferable class structure.

Fish [44]: Fish performs gradient-matching style meta-updates across domains. It is included as a lightweight DG alternative based on parameter updates that remain consistent across source domains.

CSD [36]: CSD, or Common-Specific Decomposition, separates shared and domain-specific components in the learned representation. It is included as a structured DG model for testing whether explicit decomposition helps under wave shift.

SagNet [32]: SagNet, or Style Agnostic Network, tries to separate content and style during training so that the classifier becomes less sensitive to nuisance style variation. It is included as a DG model that explicitly targets style robustness.

C.2.1.4 Domain adaptation models

Current benchmark runs evaluate UDA models with a shared MLP backbone so that differences primarily reflect the adaptation algorithm rather than backbone changes.

Domain-Adversarial Neural Networks (DANN) [9]: DANN uses adversarial training with a domain discriminator so that the representation becomes less informative about the domain label while remaining predictive for the task label. It is the standard adversarial UDA baseline.

Conditional Domain Adversarial Network (CDAN) [28]: CDAN extends DANN by conditioning the adversarial alignment on task predictions as well as learned features. It is included because conditional alignment can be stronger than marginal adversarial alignment.

Deep Adaptation Network (DAN) [26]: DAN aligns source and target features by minimizing a multi-kernel maximum mean discrepancy objective. It serves as a non-adversarial UDA baseline based on distribution matching in feature space.

DeepCORAL [47]: DeepCORAL aligns second-order statistics between source and target feature activations. It is included as a covariance-matching alternative to adversarial and MMD-based transfer methods.

Minimum Class Confusion (MCC) [19]: MCC is a non-adversarial UDA method that reduces class confusion on target predictions. It serves as a strong target-side regularization baseline.

Adversarial Discriminative Domain Adaptation (ADDA) [50]: ADDA adapts a target encoder toward a fixed source hypothesis through adversarial training. It is included as a classical two-stage UDA method.

Maximum Classifier Discrepancy (MCD) [41]: MCD uses two classifiers and explicitly maximizes and then minimizes their prediction discrepancy on target samples. It is included because classifier disagreement provides an alternative transfer signal.

Joint Adaptation Network (JAN) [27]: JAN aligns joint distributions of features and predictions through a JMMD objective. It is included as a label-aware distribution-matching UDA method.

Source Hypothesis Transfer (SHOT) [25]: SHOT is a source-free UDA method that adapts a source hypothesis to unlabeled target data without revisiting source samples. It is included because source-free transfer is relevant in practical adaptation settings.

Class-Balanced Self-Training (CBST) [57]: CBST uses iterative pseudo-labeling with class-balanced selection rules on target data. It is included as a self-training transfer baseline.

Cross-Domain Gradient Discrepancy Minimization (CGDM) [8]: CGDM is a discrepancy-based UDA method ported into the repository with a dedicated MLP-style feature extractor and dual classifiers. It provides an additional transfer baseline beyond the more standard adversarial and moment-matching methods.

Table 21: Benchmark model inventory across baseline, tabular neural, DG, and DA groups.

Group	Model
Baseline	XGB, LGB, MLP, ResNet
Tabular neural	TabNet, SAINT, TabTransformer, FTTransformer, DCN
DG	IRM, VREx, GroupDRO, MixStyle, MLDG, MASF, Fish, CSD, SagNet
UDA	DANN, CDAN, DAN, DeepCORAL, MCC, ADDA, MCD, JAN, SHOT, CBST, CGDM

C.2.2 Benchmark-Specific Modifications and External Baselines

This subsection documents specific architectural adaptations and the integration of baselines residing outside the standard DomainBed and TLL frameworks.

- **Tabular Models:** *TabNet* (via **pytorch-tabnet**), and *SAINT*, *TabTransformer*, *FTTransformer* (via **pytorch-widedeep** [38]).
- **Deep CTR & Domain Adaptation:** *DCN* is integrated from **deeppctr-torch** [43], while *MAST* and *CGDM* are ported from their original author-released implementations.
- **Advanced UDA Techniques:** *DeepCORAL*¹, *ADDA*², *MCD*³, *SHOT*⁴, and *CBST*⁵ were adapted from official public repositories to ensure performance fidelity.
- **Efficient attention for high-dimensional tabular inputs:** Tabular Transformers tokenise each input feature as a separate attention token and therefore incur a self-attention cost that scales quadratically in the number of features. Our feature tables have several thousand columns per wave (Appendix A.7.5, Table 20), which makes standard softmax self-attention memory-bound on our GPUs even at modest batch sizes. We therefore replace the attention kernel of **TabTransformer**, **SAINT**, and **FTTransformer** with a linear-attention variant [21]. Concretely, we approximate $\text{softmax}(QK^\top)V$ by $\phi(Q)(\phi(K)^\top V)$ for a shared feature map $\phi(\cdot)$, reducing the attention cost from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd^2)$ for n feature tokens. TabTransformer uses the linear-attention path exposed by its reference library (**pytorch-widedeep**); SAINT and FTTransformer are patched to route their encoder blocks through the same operator while preserving all other architectural choices. The substitution is local to the attention kernel only, so the family comparison between tabular Transformers, DG, and UDA models remains capacity-matched under a shared training protocol.

C.2.3 Unified Training Protocol

This subsection lists the values that realise the cross-method comparability controls described in Section 4.7. Every baseline, DG, and UDA method is driven by the same training loop, early-stopping contract, and model-selection rule; the only per-model differences are the HPO search spaces reported in Tables 22, 23, and 24.

¹<https://github.com/SSARCandy/DeepCORAL>

²<https://github.com/jvanvugt/pytorch-domain-adaptation>

³https://github.com/mil-tokyo/MCD_DA

⁴<https://github.com/tim-learn/SHOT>

⁵<https://github.com/yzou2/CBST>

- **Optimisation and Loss:** All neural models are trained with AdamW [29] under a cross-entropy classification loss for binary targets, with the learning rate and weight decay drawn from the HPO grid. Tree-based baselines (XGBoost, LightGBM) use their native boosted-tree objective with AUC as the evaluation metric. No explicit class-imbalance correction is applied at the loss level (`class_weight` or `pos_weight` are not used), so improvements on imbalanced targets are attributable to the objective or architecture rather than to a reweighted loss.
- **Epoch Budget and Early Stopping:** Every model is trained under a uniform maximum budget of 50 epochs. A patience-based early-stopping rule monitors validation AUROC with patience = 20, minimum improvement $\Delta = 10^{-4}$, and best-weight restoration enabled: when validation AUROC fails to improve by more than Δ for 20 consecutive epochs the run terminates and the best-AUROC checkpoint is reloaded before final evaluation. Tree-based baselines use the equivalent built-in `early_stopping_rounds = 20` on the same validation split.
- **Batch Size:** All models (baselines, DG, and UDA) are trained with a fixed batch size of 512 applied uniformly across all settings, except where this was not possible due to memory limitations (see code for details). Batch size is not treated as a tunable hyperparameter in this release: it was fixed up-front to keep the per-step optimisation budget consistent across families, since a per-model sweep over batch size would have required GPU memory and run-time budget beyond what was available for the transformer and UDA variants at the larger end of the grid.
- **Hardware:** Neural network models (MLP, ResNet, all tabular Transformers, and all DG and UDA methods) were trained on a single NVIDIA RTX PRO 5000 Blackwell GPU with 48 GB VRAM. Gradient boosted tree baselines (XGBoost, LightGBM) were trained on CPU rather than GPU, since the per tree fit does not benefit from accelerator parallelism at our feature dimensionality.
- **Validation, Checkpoint, and HPO:** Model selection is performed exclusively on the validation split defined by each benchmark’s protocol (temporal validation slice, within-wave fold-level validation partition, cross-wave source-only validation split). The selected checkpoint is the epoch with the highest validation AUROC observed during training, restored from the best-weight snapshot maintained by the early-stopping callback. HPO itself uses the same validation AUROC as its objective, with 30 Optuna [1] trials per (benchmark, label, dataset/direction, model) configuration.
- **Reproducibility:** Each run seeds `numpy.random`, `torch.manual_seed`, and `torch.cuda.manual_seed_all` at the start. Primary benchmark runs use seed 42. Split construction, HPO sampling, model initialisation, and data-loader shuffling are all seeded, so rerunning on identical hardware reproduces the reported metrics within cuDNN-level floating-point tolerance.

C.2.4 Hyperparameter Optimization Search Spaces

The comprehensive hyperparameter search spaces for all models are detailed in Tables 22–24. Our grid design follows two goals: (i) fidelity to the original implementations and (ii) a comparable optimization budget across models.

- **Adaptation for tabular data:** For DG and UDA methods with limited prior use on high dimensional longitudinal sensing tables, we set each search range starting from the corresponding reference benchmark, namely DomainBed [11] for DG methods and the Transfer-Learning-Library [18] for UDA methods, and then narrow or extend the ranges of individual hyperparameters to suit our feature dimensionality and per fold sample size.
- **Alignment with prior work:** For remaining methods, we ensure that our search spaces cover the hyperparameter ranges reported in the original papers or reference benchmarks that introduced or evaluated the method.

Table 22: Hyperparameter grids for baseline and tabular neural methods (continued in Table 23 and Table 24).

Model	Applicability	Hyperparameter	Values
Baseline Methods			
MLP / ResNet	shared	learning rate	LogUniform(10^{-5} , 10^{-2})
		weight decay	LogUniform(10^{-6} , 10^{-3})
		hidden dim	Integer[64, 512]
		dropout	Uniform[0.0, 0.5]
	MLP	num. layers	Integer[1, 8]
ResNet	num. blocks	Integer[1, 8]	
XGB	method-specific	learning rate	LogUniform(10^{-5} , 1)
		max depth	Integer[3, 10]
		min child weight	LogUniform(10^{-8} , 10^5)
		subsample	Uniform[0.5, 1.0]
		column subsample (tree / level)	Uniform[0.5, 1.0]
		gamma	{0} or LogUniform(10^{-8} , 10^2)
		λ	{0} or LogUniform(10^{-8} , 10^2)
		α	{0} or LogUniform(10^{-8} , 10^2)
		num. estimators	Integer[50, 500]
LGB	method-specific	learning rate	LogUniform(10^{-3} , 1)
		num. leaves	Integer[8, 128]
		num. estimators	Integer[50, 500]
		min child samples	Integer[5, 100]
		subsample	Uniform[0.5, 1.0]
		column subsample (tree)	Uniform[0.5, 1.0]
Tabular Neural Methods			
TabNet	method-specific	learning rate	Uniform[10^{-3} , 5×10^{-3}]
		weight decay	LogUniform(10^{-5} , 5×10^{-4})
		n_d, n_a	{8, 16}
		n_{steps}	Integer[3, 4]
		γ	Uniform[1.1, 1.5]
		λ_{sparse}	LogUniform(10^{-5} , 10^{-3})
SAINT / TabTransformer	shared	learning rate	LogUniform(2×10^{-4} , 8×10^{-4})
		weight decay	LogUniform(10^{-5} , 5×10^{-4})
		input dim	{8, 16}
		n_{heads}	{2, 4}
		n_{blocks}	Integer[1, 2]
		attn dropout	Uniform[0.0, 0.2]
		ff dropout	Uniform[0.0, 0.2]
FTTransformer	method-specific	learning rate	LogUniform(5×10^{-5} , 5×10^{-4})
		weight decay	LogUniform(10^{-5} , 5×10^{-4})
		input dim	{16, 32}
		n_{blocks}	Integer[1, 2]
		attn dropout	Uniform[0.0, 0.2]
		ff dropout	Uniform[0.0, 0.2]
		residual dropout	{0} $\cup$ Uniform[0.0, 0.1]
		ff factor	Uniform[1.0, 2.0]
DCN	method-specific	learning rate	LogUniform(5×10^{-5} , 5×10^{-3})
		weight decay	LogUniform(10^{-5} , 5×10^{-4})
		n_{hidden} layers	Integer[1, 2]
		layer size	Integer[64, 128]
		n_{cross} layers	Integer[1, 2]
		hidden dropout	Uniform[0.0, 0.2]
		cross dropout	{0} $\cup$ Uniform[0.0, 0.2]

Table 23: Hyperparameter grids for domain generalization (DG) methods (continued from Table 22; continues in Table 24).

Model	Applicability	Hyperparameter	Values
Domain Generalization Methods			
DG (shared)	all DG methods	learning rate weight decay dropout hidden dim layers	LogUniform($3 \times 10^{-5}, 3 \times 10^{-4}$) LogUniform($10^{-6}, 10^{-2}$) {0.0, 0.1, 0.5} Integer[64, 512] Integer[1, 8]
IRM	method-specific	penalty λ anneal	LogUniform($10^{-1}, 10^5$) $10^u, u \sim \text{Uniform}[0, 4]$
VREx	method-specific	penalty λ anneal	LogUniform($10^{-1}, 10^5$) $10^u, u \sim \text{Uniform}[0, 4]$
GroupDRO	method-specific	step size η	LogUniform($10^{-3}, 10^{-1}$)
MixStyle	method-specific	α p mix mode	Uniform[0.1, 0.5] Uniform[0.1, 0.9] {random, crossdomain}
MLDG	method-specific	meta step β $n_{\text{meta-test}}$	LogUniform($10^{-1}, 10$) {1, 2}
MASF	method-specific	inner lr metric lr metric weight margin temperature metric dim	LogUniform($10^{-5}, 10^{-2}$) LogUniform($10^{-5}, 10^{-2}$) LogUniform($10^{-4}, 10^{-2}$) Uniform[0.2, 2.0] Uniform[1.0, 5.0] {64, 128, 256}
Fish	method-specific	meta lr	{0.05, 0.1, 0.5}
CSD	method-specific	λ k	Uniform[0.1, 5.0] {2, 3}
SagNet	method-specific	adv weight	LogUniform($10^{-2}, 10$)

Table 24: Hyperparameter grids for unsupervised domain adaptation (UDA) methods (continued from Table 22 and Table 23).

Model	Applicability	Hyperparameter	Values
Domain Adaptation Methods			
UDA (shared)	all UDA methods	learning rate	LogUniform(10^{-5} , 10^{-2})
		weight decay	LogUniform(10^{-6} , 10^{-3})
		hidden dim	Integer[64, 512]
		layers	Integer[1, 8]
DANN	method-specific	discriminator lr	LogUniform(10^{-5} , 10^{-2})
		disc width	2^u , $u \sim \text{Uniform}[6, 10]$
		trade-off	LogUniform(10^{-2} , 10^2)
CDAN	method-specific	discriminator lr	LogUniform(10^{-5} , 10^{-2})
		disc width	2^u , $u \sim \text{Uniform}[6, 10]$
		trade-off	LogUniform(10^{-2} , 10^2)
		entropy	{T, F}
		randomized	{F, T}
DAN	method-specific	randomized dim	{256, 512, 1024}
		trade-off	LogUniform[0.1, 10.0]
		kernels	{3, 5, 7}
		linear	{F, T}
DeepCORAL	method-specific	γ	LogUniform[0.1, 10.0]
MCC	method-specific	temperature	Uniform[1.0, 5.0]
		trade-off	LogUniform(10^{-2} , 10^1)
JAN	method-specific	JMMD λ	Uniform[0.1, 10.0]
		kernels	{3, 5, 7}
		linear	{F, T}
ADDA	method-specific	discriminator lr	LogUniform(10^{-5} , 10^{-2})
		disc width	2^u , $u \sim \text{Uniform}[6, 10]$
MCD	method-specific	k (#discrepancy steps)	Integer[1, 4]
		trade-off	Uniform[0.5, 2.0]
SHOT	method-specific	cls par β_{cls}	Uniform[0.0, 1.0]
		ent par β_{ent}	Uniform[0.5, 2.0]
		label-update interval	Integer[5, 20]
		adapt epochs	Integer[10, 50]
CBST	method-specific	pretrain epochs	Integer[10, 30]
		max self-training iter	{3, 5, 7}
		init portion	Uniform[0.1, 0.3]
		portion step	Uniform[0.05, 0.2]
		max portion	Uniform[0.6, 0.9]
		retrain epochs	{3, 5, 10}
CGDM	method-specific	weight decay	LogUniform(10^{-5} , 10^{-3})

C.3 Additional Results

C.3.1 Within-User Temporal Prediction Results

The resulting AUROCs quantify the predictive signal carried by the features themselves. Across labels, gradient-boosted trees (XGBoost, LightGBM) lead in most cells of Table 25, with tabular neural models (SAINT, FTTransformer, TabTransformer) competitive on a handful of (label, dataset) pairs. The predictability of D-3-only affect labels varies significantly between classes, with the best AUROCs from 0.548 (*Depressed*) to 0.673 (*Angry*), indicating that the difficulty per-label varies substantially even within a common affect family. Across waves, arousal and valence improve monotonically from D-1 to D-3 (0.579 $\rightarrow$ 0.610 $\rightarrow$ 0.647 and 0.575 $\rightarrow$ 0.594 $\rightarrow$ 0.635), while disturbance remains near chance throughout. Stress, in contrast, is roughly stable across waves (0.636 $\rightarrow$ 0.618 $\rightarrow$ 0.646 across D-1, D-2, and D-3).

Table 25: Full within-user temporal results. Cells show test AUROC (60/20/20 chronological split per user, concatenated across users). Bold marks the best model within each row.

Label	Dataset	XGB	LGB	MLP	ResNet	TabNet	DCN	SAINT	TabTr	FTT
Shared labels										
arousal	D-1	0.579	0.564	0.522	0.520	0.505	0.510	0.502	0.503	0.517
	D-2	0.600	0.610	0.538	0.559	0.518	0.538	0.529	0.561	0.530
	D-3	0.647	0.645	0.561	0.542	0.540	0.536	0.592	0.562	0.558
disturbance	D-1	0.535	0.501	0.501	0.503	0.509	0.516	0.550	0.528	0.518
	D-2	0.587	0.575	0.494	0.511	0.481	0.518	0.560	0.550	0.532
	D-3	0.476	0.472	0.577	0.566	0.495	0.511	0.557	0.551	0.525
stress	D-1	0.618	0.636	0.559	0.555	0.500	0.535	0.502	0.558	0.513
	D-2	0.618	0.616	0.591	0.592	0.575	0.559	0.566	0.578	0.549
	D-3	0.646	0.619	0.562	0.586	0.544	0.572	0.544	0.579	0.539
valence	D-1	0.575	0.558	0.560	0.530	0.507	0.541	0.560	0.554	0.537
	D-2	0.592	0.592	0.556	0.554	0.503	0.521	0.571	0.594	0.528
	D-3	0.635	0.625	0.527	0.536	0.488	0.510	0.593	0.574	0.564
D-3 rich labels										
angry	D-3	0.673	0.633	0.537	0.544	0.492	0.502	0.533	0.520	0.515
anxious	D-3	0.483	0.515	0.560	0.548	0.516	0.511	0.575	0.556	0.540
cheerful	D-3	0.574	0.599	0.571	0.533	0.525	0.526	0.559	0.564	0.532
content	D-3	0.601	0.578	0.545	0.513	0.511	0.527	0.562	0.567	0.515
depressed	D-3	0.474	0.502	0.519	0.541	0.518	0.499	0.522	0.476	0.548
happy	D-3	0.562	0.593	0.576	0.563	0.502	0.564	0.550	0.555	0.537
relaxed	D-3	0.585	0.584	0.553	0.550	0.505	0.513	0.581	0.570	0.517
sad	D-3	0.563	0.607	0.558	0.541	0.505	0.513	0.501	0.565	0.559

C.3.2 Within-Wave Cross-User Generalization Results

We use a stratified 5-fold group split that holds out users, and report each model’s mean test AUROC together with its fold-level standard deviation. The mean shows average generalization, and the spread shows sensitivity to which users are held out. Figure 17 ranks all 29 models according to the mean within-wave AUROC for the exact mean values; refer to Table 26.

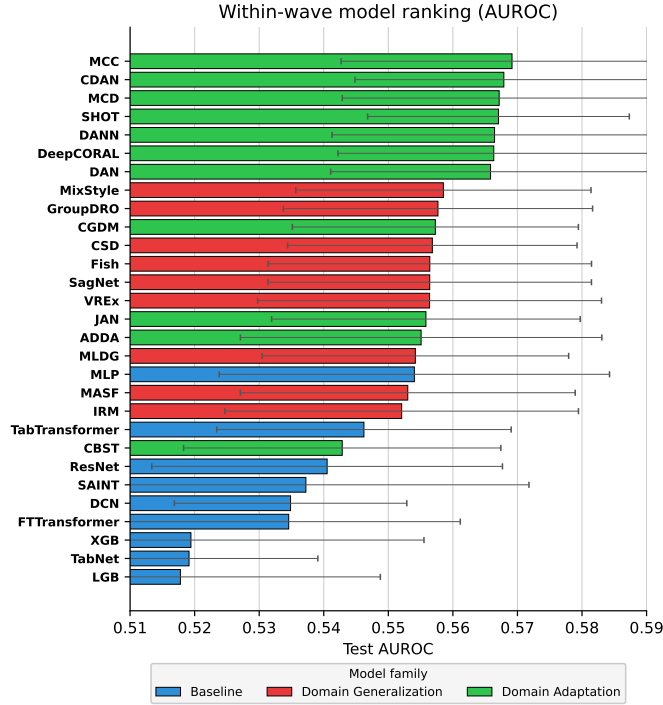


Figure 17: Within-wave model ranking by mean test AUROC. Each bar is the mean of a model’s test AUROC across all within-wave (label, dataset) rows; error bars show the standard deviation across those rows. Bars are shaded by family.

Domain adaptation dominates under cross-user shift. The best UDA model wins 11 of 12 shared-label cells. The one exception is stress on D-2, where SagNet (DG) edges out ADDA (UDA) at 0.562 vs. 0.555. UDA also wins seven of the eight D-3-only affect labels (Table 30); the one D-3 baseline win is *sad*, where XGB edges out UDA. The model-level ranking tells the same story: UDA objectives cluster at the top, DG methods just below, and baseline and tabular models at the floor. Two outliers are worth calling out. CBST is the one UDA method that drops to the bottom; we read this as its iterative pseudo-labeling needing more target data than each fold provides. The plain MLP baseline, on the other hand, sits in the middle of the DG block: when the domain is user identity, explicit DG objectives don’t beat plain ERM, but UDA methods still gain from seeing unlabeled target-user features.

DG gains little when each user is a domain. All DG methods share the MLP backbone of the MLP baseline, so any difference is attributable to the DG objective alone. In the within-wave benchmark, where each user is treated as a domain, DG objectives do not consistently improve over plain MLP on shared-label tasks. The cross-wave results below evaluate the same user-domain regularization under a stricter held-out-wave transfer setting.

Table 26: Within-wave model ranking by mean test AUROC (companion to Figure 17). Each model’s score is the mean of its per-row HPO-best test AUROC across all 20 within-wave (label, dataset) rows; $\pm$ is the standard deviation across those rows.

Rank	Model	Family	Test AUROC
1	MCC	DA	0.569 ± 0.026
2	CDAN	DA	0.568 ± 0.023
3	MCD	DA	0.567 ± 0.024
4	SHOT	DA	0.567 ± 0.020
5	DANN	DA	0.566 ± 0.025
6	DeepCORAL	DA	0.566 ± 0.024
7	DAN	DA	0.566 ± 0.025
8	MixStyle	DG	0.559 ± 0.023
9	GroupDRO	DG	0.558 ± 0.024
10	CGDM	DA	0.557 ± 0.022
11	CSD	DG	0.557 ± 0.022
12	Fish	DG	0.556 ± 0.025
13	SagNet	DG	0.556 ± 0.025
14	VREx	DG	0.556 ± 0.027
15	JAN	DA	0.556 ± 0.024
16	ADDA	DA	0.555 ± 0.028
17	MLDG	DG	0.554 ± 0.024
18	MLP	Base	0.554 ± 0.030
19	MASF	DG	0.553 ± 0.026
20	IRM	DG	0.552 ± 0.027
21	TabTransformer	Base	0.546 ± 0.023
22	CBST	DA	0.543 ± 0.025
23	ResNet	Base	0.541 ± 0.027
24	SAINT	Base	0.537 ± 0.035
25	DCN	Base	0.535 ± 0.018
26	FTTransformer	Base	0.535 ± 0.027
27	XGB	Base	0.519 ± 0.036
28	TabNet	Base	0.519 ± 0.020
29	LGB	Base	0.518 ± 0.031

Table 27: Within-wave cross-user results for baseline and tabular models. Cells show test AUROC (mean over 5 stratified user folds). Bold marks the best model per row.

Label	Dataset	XGB	LGB	MLP	ResNet	TabNet	DCN	SAINT	TabTr	FTT
Shared labels										
arousal	D-1	0.545	0.554	0.537	0.542	0.536	0.527	0.556	0.557	0.534
	D-2	0.535	0.540	0.542	0.552	0.542	0.534	0.562	0.555	0.517
	D-3	0.580	0.590	0.610	0.605	0.579	0.571	0.583	0.599	0.592
disturbance	D-1	0.495	0.517	0.537	0.506	0.506	0.525	0.521	0.542	0.519
	D-2	0.494	0.504	0.560	0.542	0.533	0.532	0.565	0.564	0.536
	D-3	0.501	0.493	0.523	0.515	0.494	0.520	0.510	0.520	0.502
stress	D-1	0.495	0.515	0.505	0.515	0.498	0.519	0.507	0.509	0.521
	D-2	0.531	0.500	0.546	0.537	0.500	0.539	0.530	0.535	0.539
	D-3	0.538	0.523	0.580	0.563	0.521	0.544	0.536	0.556	0.553
valence	D-1	0.488	0.498	0.558	0.538	0.529	0.531	0.530	0.549	0.522
	D-2	0.537	0.516	0.545	0.536	0.504	0.523	0.507	0.532	0.520
	D-3	0.556	0.551	0.576	0.555	0.530	0.538	0.566	0.571	0.527
D-3 rich labels										
angry	D-3	0.474	0.473	0.523	0.518	0.504	0.519	0.497	0.511	0.510
anxious		0.524	0.553	0.554	0.520	0.500	0.524	0.519	0.529	0.531
cheerful		0.553	0.525	0.609	0.568	0.530	0.568	0.578	0.573	0.576
content		0.533	0.540	0.594	0.577	0.521	0.568	0.583	0.561	0.578
depressed		0.441	0.462	0.518	0.516	0.508	0.514	0.532	0.530	0.504
happy		0.502	0.477	0.584	0.566	0.520	0.551	0.566	0.563	0.568
relaxed		0.484	0.512	0.556	0.548	0.508	0.539	0.550	0.544	0.542
sad		0.581	0.511	0.522	0.493	0.519	0.510	0.446	0.524	0.501

Table 28: Within-wave cross-user results for domain generalization (DG) models. Cells show test AUROC; bold marks the best model per row.

Label	Dataset	IRM	VREx	GDRO	MixStyle	MLDG	MASF	Fish	CSD	SagNet
Shared labels										
arousal	D-1	0.529	0.546	0.544	0.538	0.527	0.546	0.543	0.540	0.521
	D-2	0.567	0.562	0.567	0.563	0.556	0.562	0.565	0.566	0.556
	D-3	0.606	0.609	0.603	0.596	0.598	0.600	0.606	0.604	0.605
disturbance	D-1	0.535	0.533	0.550	0.545	0.542	0.527	0.539	0.550	0.540
	D-2	0.558	0.559	0.559	0.564	0.560	0.552	0.559	0.557	0.561
	D-3	0.523	0.537	0.537	0.536	0.532	0.532	0.534	0.535	0.535
stress	D-1	0.531	0.514	0.532	0.534	0.536	0.520	0.524	0.531	0.534
	D-2	0.536	0.554	0.559	0.557	0.554	0.555	0.553	0.555	0.562
	D-3	0.571	0.571	0.570	0.569	0.567	0.575	0.568	0.568	0.564
valence	D-1	0.542	0.547	0.551	0.543	0.538	0.545	0.550	0.551	0.536
	D-2	0.539	0.554	0.538	0.543	0.528	0.525	0.538	0.552	0.537
	D-3	0.562	0.574	0.574	0.577	0.571	0.559	0.572	0.567	0.579
D-3 rich labels										
angry	D-3	0.519	0.524	0.522	0.526	0.519	0.523	0.521	0.523	0.525
anxious		0.553	0.551	0.555	0.558	0.555	0.556	0.549	0.545	0.548
cheerful		0.594	0.607	0.605	0.605	0.601	0.598	0.606	0.604	0.603
content		0.588	0.589	0.576	0.584	0.585	0.582	0.583	0.580	0.587
depressed		0.532	0.540	0.525	0.540	0.537	0.527	0.530	0.538	0.544
happy		0.589	0.583	0.592	0.592	0.582	0.594	0.589	0.582	0.592
relaxed		0.560	0.558	0.555	0.562	0.553	0.556	0.559	0.555	0.553
sad		0.507	0.516	0.538	0.537	0.541	0.526	0.541	0.535	0.548

Table 29: Within-wave cross-user results for unsupervised domain adaptation (UDA) models. Cells show test AUROC; bold marks the best model per row.

Label	Dataset	DANN	CDAN	DAN	CORAL	MCC	ADDA	MCD	JAN	SHOT	CBST	CGDM
Shared labels												
arousal	D-1	0.570	0.561	0.531	0.569	0.566	0.552	0.576	0.562	0.566	0.534	0.566
	D-2	0.568	0.572	0.573	0.571	0.573	0.563	0.568	0.552	0.573	0.540	0.561
	D-3	0.611	0.607	0.608	0.601	0.609	0.606	0.608	0.592	0.608	0.562	0.602
disturbance	D-1	0.553	0.559	0.563	0.558	0.551	0.543	0.566	0.554	0.559	0.524	0.548
	D-2	0.558	0.566	0.563	0.566	0.564	0.555	0.561	0.560	0.567	0.538	0.545
	D-3	0.538	0.544	0.546	0.534	0.555	0.525	0.537	0.518	0.550	0.508	0.526
stress	D-1	0.503	0.513	0.516	0.512	0.503	0.508	0.510	0.508	0.538	0.505	0.506
	D-2	0.548	0.550	0.549	0.541	0.548	0.555	0.544	0.516	0.552	0.550	0.548
	D-3	0.579	0.579	0.581	0.575	0.572	0.564	0.573	0.551	0.573	0.575	0.557
valence	D-1	0.570	0.570	0.563	0.564	0.567	0.539	0.565	0.555	0.565	0.532	0.553
	D-2	0.573	0.573	0.570	0.566	0.575	0.573	0.573	0.567	0.563	0.523	0.552
	D-3	0.581	0.575	0.578	0.579	0.586	0.532	0.586	0.563	0.577	0.536	0.573
D-3 rich labels												
angry	D-3	0.539	0.544	0.543	0.539	0.533	0.522	0.542	0.536	0.542	0.512	0.540
anxious		0.560	0.562	0.548	0.565	0.559	0.559	0.557	0.558	0.556	0.536	0.554
cheerful		0.599	0.603	0.611	0.609	0.613	0.601	0.607	0.590	0.608	0.584	0.591
content		0.592	0.600	0.593	0.593	0.598	0.593	0.598	0.574	0.587	0.594	0.578
depressed		0.557	0.553	0.556	0.553	0.556	0.529	0.557	0.543	0.547	0.537	0.549
happy		0.604	0.601	0.600	0.604	0.609	0.599	0.596	0.597	0.597	0.571	0.587
relaxed		0.569	0.571	0.572	0.570	0.574	0.547	0.567	0.569	0.566	0.557	0.564
sad		0.556	0.556	0.552	0.558	0.571	0.535	0.555	0.550	0.548	0.539	0.545

Table 30: Within-wave family-best comparison. Each cell reports the best-performing model within a family as *Model: AUROC ($\pm$ fold std)*; bold marks the best family per row.

Label	Dataset	Baseline	DG	DA
Shared labels				
arousal	D-1	TabTr: 0.557 ($\pm$ 0.016)	MASF: 0.546 ($\pm$ 0.013)	MCD: 0.576 ($\pm$0.013)
	D-2	SAINT: 0.562 ($\pm$ 0.025)	GDRO: 0.567 ($\pm$ 0.016)	SHOT: 0.573 ($\pm$0.015)
	D-3	MLP: 0.610 ($\pm$ 0.007)	VREx: 0.609 ($\pm$ 0.005)	DANN: 0.611 ($\pm$0.003)
disturbance	D-1	TabTr: 0.542 ($\pm$ 0.011)	CSD: 0.550 ($\pm$ 0.018)	MCD: 0.566 ($\pm$0.029)
	D-2	SAINT: 0.565 ($\pm$ 0.014)	MixStyle: 0.564 ($\pm$ 0.011)	SHOT: 0.567 ($\pm$0.008)
	D-3	MLP: 0.523 ($\pm$ 0.011)	GDRO: 0.537 ($\pm$ 0.013)	MCC: 0.555 ($\pm$0.023)
stress	D-1	FTT: 0.521 ($\pm$ 0.020)	MLDG: 0.536 ($\pm$ 0.008)	SHOT: 0.538 ($\pm$0.017)
	D-2	MLP: 0.546 ($\pm$ 0.015)	SagNet: 0.562 ($\pm$0.016)	ADDA: 0.555 ($\pm$ 0.019)
	D-3	MLP: 0.580 ($\pm$ 0.016)	MASF: 0.575 ($\pm$ 0.014)	DAN: 0.581 ($\pm$0.018)
valence	D-1	MLP: 0.558 ($\pm$ 0.019)	GDRO: 0.551 ($\pm$ 0.031)	CDAN: 0.570 ($\pm$0.015)
	D-2	MLP: 0.545 ($\pm$ 0.013)	VREx: 0.554 ($\pm$ 0.023)	MCC: 0.575 ($\pm$0.028)
	D-3	MLP: 0.576 ($\pm$ 0.019)	SagNet: 0.579 ($\pm$ 0.021)	MCC: 0.586 ($\pm$0.014)
D-3 rich labels				
angry	D-3	MLP: 0.523 ($\pm$ 0.019)	MixStyle: 0.526 ($\pm$ 0.022)	CDAN: 0.544 ($\pm$0.026)
anxious		MLP: 0.554 ($\pm$ 0.016)	MixStyle: 0.558 ($\pm$ 0.020)	CORAL: 0.565 ($\pm$0.011)
cheerful		MLP: 0.609 ($\pm$ 0.025)	VREx: 0.607 ($\pm$ 0.025)	MCC: 0.613 ($\pm$0.032)
content		MLP: 0.594 ($\pm$ 0.012)	VREx: 0.589 ($\pm$ 0.013)	CDAN: 0.600 ($\pm$0.013)
depressed		SAINT: 0.532 ($\pm$ 0.043)	SagNet: 0.544 ($\pm$ 0.024)	DANN: 0.557 ($\pm$0.010)
happy		MLP: 0.584 ($\pm$ 0.023)	MASF: 0.594 ($\pm$ 0.013)	MCC: 0.609 ($\pm$0.012)
relaxed		MLP: 0.556 ($\pm$ 0.013)	MixStyle: 0.562 ($\pm$ 0.013)	MCC: 0.574 ($\pm$0.026)
sad		XGB: 0.581 ($\pm$0.059)	SagNet: 0.548 ($\pm$ 0.053)	MCC: 0.571 ($\pm$ 0.024)

C.3.3 Cross-Wave Generalization Results

Tables 31–33 report per-direction results for baseline, DG, and UDA models across the six single-source and three multi-source directions. Family-level rankings are substantially reordered. DG achieves the best mean under single-source transfer (0.542 ± 0.017), the baseline family leads under multi-source pooling (0.551 ± 0.015) and UDA trails in both (0.531 single, 0.532 multi). Multi-source pooling yields modest, label-dependent gains over single-source transfer (+0.004 AUROC for arousal, +0.006 for valence, +0.005 for disturbance, and +0.005 for stress), and most of this improvement accrues to baseline models rather than to methods that explicitly model domain structure. Direction asymmetries are most significant for arousal involving D-3, with D-2 $\rightarrow$ D-3 exceeding D-3 $\rightarrow$ D-2 by 0.035 AUROC and D-1 $\rightarrow$ D-3 exceeding D-3 $\rightarrow$ D-1 by 0.023, whereas the remaining shared labels are markedly more symmetric (asymmetry below 0.02 AUROC for valence and stress and below 0.01 for disturbance).

Table 31: Full cross-wave AUROC results for baseline and tabular models. Best per row in **bold**.

Label	Train $\rightarrow$ Test	XGB	LGB	MLP	ResNet	TabNet	DCN	SAINT	TabTr	FTT
arousal	D-1 $\rightarrow$ D-2	0.535	0.529	0.498	0.494	0.500	0.531	0.538	0.510	0.538
	D-1 $\rightarrow$ D-3	0.545	0.572	0.520	0.535	0.499	0.540	0.545	0.515	0.545
	D-2 $\rightarrow$ D-1	0.538	0.528	0.512	0.534	0.510	0.525	0.553	0.517	0.553
	D-2 $\rightarrow$ D-3	0.582	0.565	0.539	0.554	0.507	0.566	0.580	0.535	0.580
	D-3 $\rightarrow$ D-1	0.548	0.541	0.545	0.537	0.515	0.538	0.552	0.533	0.552
	D-3 $\rightarrow$ D-2	0.529	0.541	0.527	0.514	0.510	0.543	0.533	0.527	0.533
	D-1+D-2 $\rightarrow$ D-3	0.578	0.577	0.537	0.552	0.504	0.551	0.562	0.536	0.562
	D-1+D-3 $\rightarrow$ D-2	0.569	0.571	0.529	0.541	0.523	0.526	0.538	0.537	0.538
disturbance	D-2+D-3 $\rightarrow$ D-1	0.532	0.542	0.527	0.531	0.525	0.523	0.539	0.520	0.539
	D-1 $\rightarrow$ D-2	0.518	0.512	0.552	0.527	0.499	0.526	0.540	0.553	0.540
	D-1 $\rightarrow$ D-3	0.515	0.517	0.525	0.502	0.501	0.541	0.517	0.513	0.517
	D-2 $\rightarrow$ D-1	0.528	0.529	0.536	0.560	0.500	0.527	0.552	0.547	0.552
	D-2 $\rightarrow$ D-3	0.515	0.498	0.530	0.525	0.501	0.542	0.532	0.520	0.532
	D-3 $\rightarrow$ D-1	0.541	0.523	0.513	0.508	0.490	0.527	0.508	0.516	0.508
	D-3 $\rightarrow$ D-2	0.538	0.533	0.506	0.519	0.498	0.526	0.512	0.504	0.512
	D-1+D-2 $\rightarrow$ D-3	0.537	0.518	0.525	0.523	0.497	0.542	0.520	0.524	0.520
stress	D-1+D-3 $\rightarrow$ D-2	0.565	0.550	0.536	0.536	0.510	0.527	0.540	0.529	0.540
	D-2+D-3 $\rightarrow$ D-1	0.559	0.561	0.536	0.534	0.507	0.527	0.534	0.534	0.534
	D-1 $\rightarrow$ D-2	0.525	0.524	0.533	0.517	0.497	0.502	0.494	0.540	0.522
	D-1 $\rightarrow$ D-3	0.511	0.500	0.507	0.502	0.497	0.489	0.493	0.525	0.518
	D-2 $\rightarrow$ D-1	0.532	0.526	0.531	0.532	0.511	0.513	0.506	0.500	0.523
	D-2 $\rightarrow$ D-3	0.524	0.534	0.504	0.502	0.505	0.527	0.492	0.509	0.532
	D-3 $\rightarrow$ D-1	0.504	0.530	0.518	0.515	0.502	0.505	0.499	0.492	0.508
	D-3 $\rightarrow$ D-2	0.519	0.524	0.542	0.540	0.503	0.509	0.522	0.526	0.518
valence	D-1+D-2 $\rightarrow$ D-3	0.522	0.520	0.512	0.512	0.491	0.522	0.491	0.493	0.508
	D-1+D-3 $\rightarrow$ D-2	0.532	0.531	0.518	0.513	0.505	0.523	0.512	0.516	0.513
	D-2+D-3 $\rightarrow$ D-1	0.528	0.535	0.536	0.521	0.507	0.519	0.509	0.493	0.498
	D-1 $\rightarrow$ D-2	0.521	0.514	0.512	0.504	0.505	0.507	0.534	0.515	0.534
	D-1 $\rightarrow$ D-3	0.551	0.533	0.530	0.531	0.499	0.541	0.523	0.536	0.523
	D-2 $\rightarrow$ D-1	0.530	0.509	0.514	0.516	0.506	0.518	0.541	0.512	0.541
	D-2 $\rightarrow$ D-3	0.542	0.529	0.535	0.519	0.505	0.541	0.528	0.526	0.528
	D-3 $\rightarrow$ D-1	0.527	0.561	0.522	0.516	0.508	0.520	0.525	0.534	0.525
	D-3 $\rightarrow$ D-2	0.528	0.536	0.536	0.540	0.507	0.507	0.545	0.546	0.545
	D-1+D-2 $\rightarrow$ D-3	0.543	0.562	0.527	0.531	0.499	0.540	0.532	0.524	0.532
	D-1+D-3 $\rightarrow$ D-2	0.553	0.556	0.521	0.527	0.512	0.507	0.537	0.520	0.537
	D-2+D-3 $\rightarrow$ D-1	0.535	0.559	0.514	0.523	0.528	0.520	0.546	0.520	0.546

Table 32: Full cross-wave AUROC results for DG models. Best per row in **bold**.

Label	Train → Test	IRM	VREx	GDRO	MixStyle	MLDG	MASF	Fish	CSD	SagNet
arousal	D-1 → D-2	0.565	0.544	0.554	0.553	0.559	0.560	0.556	0.547	0.533
	D-1 → D-3	0.582	0.532	0.564	0.561	0.583	0.561	0.574	0.558	0.557
	D-2 → D-1	0.544	0.546	0.542	0.543	0.533	0.531	0.544	0.548	0.540
	D-2 → D-3	0.595	0.595	0.590	0.591	0.570	0.585	0.593	0.593	0.581
	D-3 → D-1	0.550	0.550	0.550	0.552	0.553	0.559	0.554	0.560	0.547
	D-3 → D-2	0.556	0.556	0.555	0.560	0.557	0.552	0.558	0.559	0.550
	D-1+D-2 → D-3	0.590	0.589	0.568	0.584	0.575	0.586	0.587	0.591	0.558
	D-1+D-3 → D-2	0.552	0.555	0.545	0.553	0.545	0.535	0.553	0.552	0.546
	D-2+D-3 → D-1	0.542	0.541	0.547	0.531	0.552	0.544	0.544	0.533	0.532
disturbance	D-1 → D-2	0.545	0.549	0.552	0.555	0.553	0.546	0.551	0.548	0.556
	D-1 → D-3	0.535	0.527	0.529	0.529	0.532	0.528	0.525	0.537	0.530
	D-2 → D-1	0.545	0.545	0.541	0.546	0.541	0.536	0.546	0.549	0.524
	D-2 → D-3	0.524	0.524	0.530	0.529	0.532	0.513	0.530	0.535	0.507
	D-3 → D-1	0.535	0.534	0.529	0.532	0.531	0.533	0.529	0.536	0.515
	D-3 → D-2	0.529	0.529	0.522	0.533	0.523	0.517	0.522	0.525	0.512
	D-1+D-2 → D-3	0.524	0.525	0.518	0.525	0.519	0.514	0.513	0.521	0.499
	D-1+D-3 → D-2	0.543	0.543	0.539	0.542	0.540	0.544	0.540	0.553	0.535
	D-2+D-3 → D-1	0.540	0.524	0.525	0.527	0.536	0.532	0.525	0.525	0.534
stress	D-1 → D-2	0.535	0.529	0.538	0.526	0.531	0.531	0.527	0.520	0.521
	D-1 → D-3	0.503	0.511	0.509	0.504	0.507	0.508	0.502	0.503	0.499
	D-2 → D-1	0.527	0.527	0.532	0.526	0.529	0.529	0.532	0.531	0.534
	D-2 → D-3	0.516	0.514	0.512	0.509	0.511	0.510	0.518	0.515	0.505
	D-3 → D-1	0.510	0.511	0.515	0.512	0.503	0.524	0.512	0.512	0.517
	D-3 → D-2	0.534	0.537	0.542	0.539	0.542	0.547	0.541	0.545	0.545
	D-1+D-2 → D-3	0.501	0.508	0.495	0.513	0.500	0.514	0.508	0.509	0.511
	D-1+D-3 → D-2	0.530	0.527	0.529	0.532	0.523	0.531	0.528	0.538	0.519
	D-2+D-3 → D-1	0.539	0.537	0.537	0.537	0.536	0.532	0.536	0.543	0.538
valence	D-1 → D-2	0.536	0.536	0.544	0.540	0.539	0.528	0.533	0.535	0.540
	D-1 → D-3	0.539	0.539	0.540	0.541	0.534	0.533	0.536	0.534	0.536
	D-2 → D-1	0.539	0.540	0.522	0.540	0.539	0.530	0.537	0.537	0.521
	D-2 → D-3	0.551	0.551	0.542	0.552	0.554	0.543	0.548	0.550	0.528
	D-3 → D-1	0.529	0.530	0.533	0.527	0.534	0.535	0.531	0.537	0.532
	D-3 → D-2	0.543	0.543	0.542	0.540	0.541	0.552	0.538	0.542	0.539
	D-1+D-2 → D-3	0.547	0.540	0.545	0.552	0.550	0.533	0.553	0.528	0.530
	D-1+D-3 → D-2	0.536	0.531	0.544	0.542	0.528	0.549	0.549	0.548	0.539
	D-2+D-3 → D-1	0.541	0.540	0.522	0.527	0.533	0.532	0.542	0.526	0.537

Table 33: Full cross-wave AUROC results for UDA models. Best per row in **bold**. CGDM was not evaluated in the cross-wave benchmark in this release.

Label	Train $\rightarrow$ Test	DANN	CDAN	DAN	CORAL	MCC	ADDA	MCD	JAN	SHOT	CBST
arousal	D-1 $\rightarrow$ D-2	0.535	0.547	0.536	0.498	0.538	0.511	0.497	0.544	0.479	0.481
	D-1 $\rightarrow$ D-3	0.477	0.546	0.515	0.530	0.475	0.531	0.499	0.548	0.518	0.508
	D-2 $\rightarrow$ D-1	0.525	0.521	0.518	0.528	0.504	0.518	0.499	0.554	0.508	0.526
	D-2 $\rightarrow$ D-3	0.533	0.526	0.545	0.512	0.503	0.517	0.522	0.546	0.540	0.543
	D-3 $\rightarrow$ D-1	0.558	0.535	0.551	0.541	0.512	0.538	0.492	0.537	0.554	0.551
	D-3 $\rightarrow$ D-2	0.527	0.498	0.520	0.541	0.535	0.518	0.506	0.513	0.511	0.523
	D-1+D-2 $\rightarrow$ D-3	0.490	0.533	0.536	0.545	0.504	0.499	0.523	0.547	0.552	0.498
	D-1+D-3 $\rightarrow$ D-2	0.532	0.530	0.531	0.522	0.523	0.520	0.512	0.502	0.515	0.493
disturbance	D-2+D-3 $\rightarrow$ D-1	0.521	0.534	0.536	0.549	0.512	0.540	0.490	0.551	0.520	0.532
	D-1 $\rightarrow$ D-2	0.534	0.528	0.549	0.533	0.511	0.544	0.507	0.515	0.542	0.516
	D-1 $\rightarrow$ D-3	0.512	0.519	0.522	0.503	0.504	0.529	0.501	0.487	0.516	0.513
	D-2 $\rightarrow$ D-1	0.537	0.538	0.537	0.546	0.498	0.551	0.499	0.508	0.542	0.510
	D-2 $\rightarrow$ D-3	0.528	0.535	0.529	0.520	0.502	0.507	0.504	0.492	0.526	0.502
	D-3 $\rightarrow$ D-1	0.532	0.523	0.521	0.509	0.497	0.508	0.489	0.509	0.501	0.500
	D-3 $\rightarrow$ D-2	0.493	0.514	0.534	0.521	0.515	0.514	0.529	0.480	0.488	0.467
	D-1+D-2 $\rightarrow$ D-3	0.508	0.533	0.515	0.527	0.510	0.521	0.493	0.513	0.528	0.502
stress	D-1+D-3 $\rightarrow$ D-2	0.545	0.531	0.547	0.529	0.523	0.518	0.533	0.540	0.546	0.541
	D-2+D-3 $\rightarrow$ D-1	0.530	0.536	0.533	0.535	0.529	0.499	0.520	0.502	0.535	0.529
	D-1 $\rightarrow$ D-2	0.500	0.517	0.512	0.512	0.477	0.529	0.494	0.531	0.533	0.534
	D-1 $\rightarrow$ D-3	0.490	0.497	0.510	0.503	0.505	0.510	0.513	0.511	0.506	0.511
	D-2 $\rightarrow$ D-1	0.506	0.535	0.537	0.529	0.505	0.533	0.507	0.517	0.535	0.535
	D-2 $\rightarrow$ D-3	0.489	0.510	0.502	0.507	0.511	0.508	0.506	0.515	0.506	0.502
	D-3 $\rightarrow$ D-1	0.517	0.515	0.499	0.512	0.503	0.515	0.498	0.526	0.509	0.507
	D-3 $\rightarrow$ D-2	0.538	0.544	0.542	0.542	0.540	0.540	0.506	0.529	0.541	0.545
valence	D-1+D-2 $\rightarrow$ D-3	0.510	0.505	0.512	0.513	0.499	0.516	0.506	0.497	0.512	0.503
	D-1+D-3 $\rightarrow$ D-2	0.534	0.536	0.531	0.527	0.520	0.508	0.500	0.544	0.514	0.525
	D-2+D-3 $\rightarrow$ D-1	0.539	0.539	0.535	0.536	0.509	0.530	0.516	0.527	0.538	0.531
	D-1 $\rightarrow$ D-2	0.556	0.547	0.561	0.529	0.533	0.534	0.501	0.508	0.506	0.494
	D-1 $\rightarrow$ D-3	0.503	0.502	0.531	0.545	0.532	0.517	0.512	0.538	0.527	0.517
	D-2 $\rightarrow$ D-1	0.536	0.522	0.514	0.541	0.522	0.518	0.513	0.538	0.510	0.517
	D-2 $\rightarrow$ D-3	0.511	0.514	0.523	0.519	0.499	0.509	0.484	0.501	0.535	0.516
	D-3 $\rightarrow$ D-1	0.528	0.501	0.508	0.501	0.497	0.502	0.510	0.553	0.527	0.528
	D-3 $\rightarrow$ D-2	0.528	0.519	0.525	0.522	0.490	0.545	0.507	0.468	0.523	0.506
	D-1+D-2 $\rightarrow$ D-3	0.509	0.517	0.522	0.532	0.499	0.489	0.508	0.522	0.520	0.513
	D-1+D-3 $\rightarrow$ D-2	0.520	0.528	0.522	0.512	0.535	0.515	0.510	0.522	0.528	0.516
	D-2+D-3 $\rightarrow$ D-1	0.537	0.521	0.517	0.524	0.497	0.507	0.499	0.504	0.516	0.508

Table 34: Cross-wave family-best comparison by generalization setting. Each cell reports the best-performing model within a family as *Model*: *AUROC* (averaged across directions in that setting). Bold marks the best family per (label, setting).

Label	Single-source ($n = 6$)			Multi-source ($n = 3$)		
	Baseline	DG	DA	Baseline	DG	DA
arousal	FTT: 0.550	IRM: 0.565	JAN: 0.540	LGB: 0.563	VREx: 0.561	CORAL: 0.539
disturbance	DCN: 0.531	CSD: 0.538	DAN: 0.532	XGB: 0.554	IRM: 0.536	SHOT: 0.536
stress	LGB: 0.523	MASF: 0.525	ADDA: 0.523	LGB: 0.529	CSD: 0.530	DANN: 0.528
valence	XGB: 0.533	MLDG: 0.540	DAN: 0.527	LGB: 0.559	Fish: 0.548	CORAL: 0.523
Mean $\pm$ Std	0.534 $\pm$ 0.011	0.542 $\pm$ 0.017	0.531 $\pm$ 0.008	0.551 $\pm$ 0.015	0.544 $\pm$ 0.014	0.532 $\pm$ 0.007

Table 35: Cross-wave model ranking by mean test AUROC. Each models score is the mean of its per-row test AUROC across all 36 cross-wave (label, transfer-direction) cells; $\pm$ is the standard deviation across those rows.

Rank	Model	Family	Test AUROC
1	IRM	DG	0.540 ± 0.019
2	CSD	DG	0.540 ± 0.018
3	Fish	DG	0.539 ± 0.019
4	MixStyle	DG	0.538 ± 0.018
5	MLDG	DG	0.538 ± 0.017
6	VREx	DG	0.537 ± 0.018
7	GDRO	DG	0.537 ± 0.017
8	MASF	DG	0.536 ± 0.018
9	XGB	Base	0.535 ± 0.020
10	LGB	Base	0.533 ± 0.023
11	FTT	Base	0.531 ± 0.017
12	SAINT	Base	0.531 ± 0.017
13	SagNet	DG	0.531 ± 0.018
14	DCN	Base	0.528 ± 0.013
15	DAN	DA	0.526 ± 0.014
16	ResNet	Base	0.526 ± 0.014
17	MLP	Base	0.524 ± 0.012
18	CORAL	DA	0.524 ± 0.014
19	CDAN	DA	0.523 ± 0.013
20	TabTr	Base	0.523 ± 0.012
21	DANN	DA	0.522 ± 0.017
22	SHOT	DA	0.521 ± 0.016
23	JAN	DA	0.520 ± 0.021
24	ADDA	DA	0.518 ± 0.014
25	CBST	DA	0.513 ± 0.016
26	MCC	DA	0.511 ± 0.016
27	MCD	DA	0.508 ± 0.011
28	TabNet	Base	0.505 ± 0.008

D Data Hosting, Licensing, Ethics, and Governance

D.1 Data Hosting, Licensing, and Maintenance Plan

Host. The dataset is hosted on Harvard Dataverse at <https://doi.org/10.7910/DVN/QGRVU8>. Due to the sensitive nature of the data – including physiological signals, location traces, and communication metadata – the dataset is released under open credentialed access. Researchers must agree to the data use agreement before downloading any files. This is the dataset webpage <https://emophone.iclab.dev/>.

Code repository. All benchmark code, preprocessing scripts, EDA notebooks, and dataset documentation are openly available at <https://github.com/Kaist-ICLab/EmoPhone>.

License and access terms. The benchmark code is released under Apache-2.0. Dataset documentation, metadata, and supporting materials are released under CC BY 4.0. Because the dataset contains sensitive human-subject mobile and wearable sensing data, the data files are released under credentialed access and governed by a dataset-specific Data Use Agreement. In the Dataverse record, CC BY 4.0 applies to the documentation and metadata, whereas access to restricted data files is controlled by the credentialed-access terms and Data Use Agreement, which prohibit re-identification attempts, redistribution to unauthorized parties, and non-research misuse.

Repository structure. The repository is organised as follows:

```
CrossUserDataset/  
├── README.md .....  
├── DATASHEET.md .....  
├── LICENSE .....  
├── CITATION.cff .....  
├── metadata/  
│   └── croissant.json .....  
├── data/  
│   ├── README.md .....  
│   └── schema.md .....  
├── preprocessing/  
│   ├── README.md .....  
│   └── pipeline_decisions.md .....  
├── benchmark/  
│   ├── README.md .....  
│   ├── utils/  
│   ├── setting_a/  
│   ├── setting_b/  
│   └── setting_c/  
└── eda/  
    └── README.md .....
```

Long-term preservation. The dataset will be maintained by the authorship team via Harvard Dataverse, which provides DOI-based persistent access. The GitHub repository will be actively maintained alongside the dataset, with updates applied as benchmark code evolves. Error corrections and documentation updates will be versioned and communicated through the repository’s release notes.

D.2 Author Responsibility Statement

All authors collectively bear full responsibility for the content of this paper, the EmoPhone dataset, the benchmark code, and all associated materials, including the accuracy of reported results, the completeness of dataset documentation, and the validity of the benchmark design and evaluation protocol.

We confirm that the dataset does not contain content that violates privacy rights, intellectual property rights, or other legal rights. All raw data were processed with privacy-preserving transformations

prior to release: contact numbers in call and message logs were one-way hashed using MD5, Wi-Fi BSSID and Bluetooth MAC addresses were replaced with UUIDs, and GPS longitude values were randomly displaced per participant to preserve relative behavioral structure without exposing precise locations. Privacy-sensitive identifiers were transformed and access is credentialed.

We confirm that all three waves of data collection (D-1, D-2, and D-3, conducted in 2020, 2021, and 2022 respectively) were conducted under protocols approved by the relevant Institutional Review Board (IRB). All participants were informed of the study’s purpose, procedures, and data usage prior to enrollment, and provided written informed consent before participation. The consent form, including a clause on voluntary public dataset release, is included in the supplementary materials.

We are committed to the long-term availability and maintenance of both the dataset and the benchmark code. The dataset and benchmark code will be maintained by the authorship team as described in Section D.1, with versioned updates communicated through the repository’s release notes.

D.3 Ethics and Governance

All data collection procedures were conducted under institutional review and informed consent. Participants joined voluntarily, received compensation, and retained the right to withdraw. To improve transparency, we include an English translation of the participant consent form in the supplementary materials. The release is intended to support responsible academic research in affective computing, with clear attention to participant protection and appropriate limits on downstream use.

D.4 Privacy Protection and Release

The release follows privacy-preserving transformations at the raw-data level. Contact numbers in call and message logs were one-way hashed, Wi-Fi and Bluetooth identifiers were replaced with UUIDs, and GPS longitude was randomly displaced per participant so that relative behavioral structure is preserved without exposing precise locations. Quality-control filtering was also applied before modeling and release, including the removal of structurally invalid rows and exclusion of data streams with inadequate collection rates.